An Introduction to Inverse Problems

CIMPA-ICTP Courses

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Outline

Acknowledgements

This lecture is based on the course on inverse problems taught by Dr. Qinian Jin at The Australian National University.

Lecture notes are linked on my website:
`https://sites.google.com/view/mathisreal/home`.

1. Examples of inverse problems

Question: *What is the definition of inverse problems?*

There is no universal formal definition. J. B. Keller gave the following description (J. B. Keller. Inverse problems. Amer. Math. Monthly, 83:107–118, 1976.):

"We call two problems inverses of one another if the formulation of each involves all or part of the solution of the other. Often, for historical reasons, one of the two problems has been studied extensively for some time, while the other is newer and not so well understood. In such cases, the former problem is called the direct problem, while the latter is called the inverse problem."

Definition 1.1

A problem is called well-posed in the sense of Hadamard if

- the problem have a solution; and
- it has a unique solution, and
- the solution depends continuously on the data.

If one of the requirements is violated, the problem is called ill-posed.

If two problems are inverse to each other, we will call the well-posed one as direct problem, and the ill-posed one as inverse problem.

Example 1. (Numerical differentiation)

Integration and differentiation are two problems inverse to each other. The intergration problem is to find

$$\int_0^x f(\xi)d\xi$$

for a given function $f(x)$. The differentiation problem is to determine the derivative $\varphi(x) := f'$ for a given function f; it can be reformulated as the equation

$$\int_0^x \varphi(\xi)d\xi = f(x) - f(0).$$

it is easy to see that integration is a well-posed problem. However, differentiation is in general ill-posed.

For instance, let f be a continuous function on $[-1,1]$. For any small $\delta > 0$, consider the function

$$f_\delta(x) = f(x) + \delta \sin\left(\frac{x}{\delta^2}\right)$$

Then

$$\|f - f_\delta\|_\infty := \max_{x\in[-1,1]} |f(x) - f_\delta(x)| = \delta$$

and

$$\|f_\delta'(x) - f'(x)\|_\infty = \frac{1}{\delta} \max_{x\in[-1,1]} \left|\cos\left(\frac{x}{\delta^2}\right)\right| = \frac{1}{\delta}$$

Therefore, we can make f_δ as close to f as we want, but the error between f_δ' and f' can be arbitrary large. Thus, differentiation is ill-posed and we will call it an inverse problem.

If we don't know f but only know f_δ and the error level δ, can we find an approximation of f'?

To this end, for any small $h > 0$ we introduce

$$R_h f_\delta(x) := \frac{f_\delta(x+h) - f_\delta(x-h)}{2h}$$

which is the central difference quotient of f_δ. We consider the approximation property of $R_h f_\delta$ to f'.

We have

$$
\begin{aligned}
R_h f_\delta(x) - f'(x) =& \frac{f_\delta(x+h) - f_\delta(x-h)}{2h} - f'(x) \\
=& \frac{f_\delta(x+h) - f(x+h) - f_\delta(x-h) - f(x+h)}{2h} \\
&+ \frac{f(x+h) - f(x-h)}{2h} - f'(x)
\end{aligned}
$$

Therefore

$$
|R_h f_\delta(x) - f'(x)| \leq \frac{\delta}{h} + \left| \frac{f(x+h) - f(x-h)}{2h} - f'(x) \right|.
$$

In order to proceed further, we need some a priori information on the smoothness of f'.

We assume first that we know $f \in C^2$ (a priori information). Then by Taylor's expansion

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(\xi)h^3$$
$$f(x-h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{1}{6}f'''(\eta)h^3$$

where $x \leq \xi \leq x+h$ and $x-h \leq \eta \leq x$. Thus

$$\frac{f(x+h) - f(x-h)}{2h} - f'(x) = \frac{1}{12}(f'''(\xi) + f'''(\eta))h^2.$$

Consequently

$$|R_h f_\delta(x) - f'(x)| \leq \frac{\delta}{h} + \frac{1}{6}\left\|f'''\right\|_\infty h^2.$$

If we choose $h \sim \delta^{1/3}$, then we obtain

$$|R_h f_\delta(x) - f'(x)| \leq O(\delta^{2/3}).$$

Next we assume $f' \in C^\alpha$ for some $0 < \alpha \leq 1$, i.e. there is a constant C_f depending on f' such that

$$|f'(x_1) - f'(x_2)| \leq C_f \left| x_1 - x_2 \right|^\alpha, \quad \forall x_1, x_2,$$

and we say that f' is Hölder continuous with exponent α. By the fundamental theorem of calculus, we have

$$\begin{aligned} \frac{f(x+h) - f(x-h)}{2h} - f'(x) &= \frac{1}{2h} \int_{x-h}^{x+h} f'(\xi) d\xi - f'(x) \\ &= \frac{1}{2h} \int_{x-h}^{x+h} [f'(\xi) - f'(x)] d\xi. \end{aligned}$$

By the condition $f' \in C^\alpha$ we obtain

$$\begin{aligned}\left|\frac{f(x+h)-f(x-h)}{2h}-f'(x)\right| &\le \frac{1}{2h}\int_{x-h}^{x+h}|f'(\xi)-f'(x)|\,d\xi\\ &\le \frac{C_f}{2h}\int_{x-h}^{x+h}|\xi-x|^\alpha\,d\xi\\ &= \frac{C_f}{1+\alpha}h^\alpha\end{aligned}$$

Therefore

$$|R_h f_\delta(x)-f'(x)| \le \frac{\delta}{h}+\frac{C_f}{1+\alpha}h^\alpha.$$

By taking $h \sim \delta^{1/(1+\alpha)}$ we obtain

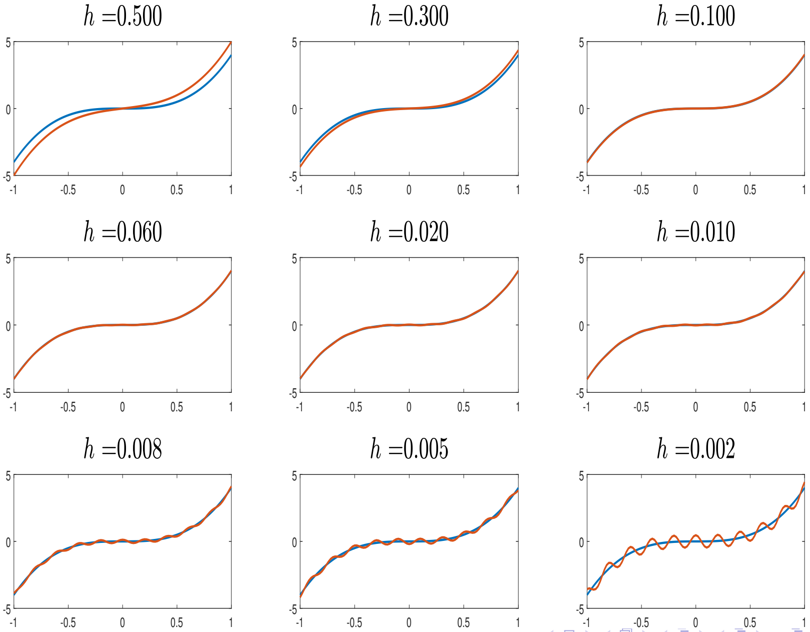
$$|R_h f_\delta(x)-f'(x)| \le O(\delta^{\alpha/(1+\alpha)}).$$

Numerical simulations

Simulation 1. Let $f(x) = x^4$ whose derivative is $f'(x) = 4x^3$. Assume we only know f_δ satisfying $\|f_\delta - f\|_\infty = \delta$, we use $R_h f_\delta(x)$ to approximate $f'(x)$. In the first figure, we take $\delta = 0.001$ and choose several different values of h to see the approximation property.

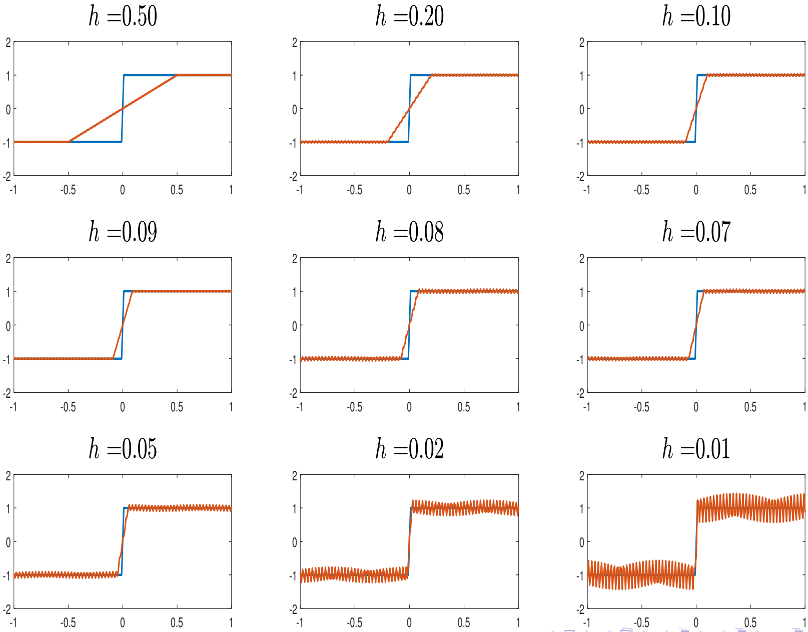
Simulation 2. Let $f(x) = |x|$ whose derivative is $f'(x) = \text{sign}(x)$. Assume we only know f_δ satisfying $\|f_\delta - f\|_\infty = \delta$, we use $R_h f_\delta(x)$ to approximate $f'(x)$. In the first figure, we take $\delta = 0.005$ and choose several different values of h to see the approximation property.

Example 1



h =0.500
h =0.300
h =0.100
h =0.060
h =0.020
h =0.010
h =0.008
h =0.005
h =0.002
5
0
-5
-1
-0.5
0
0.5
1

Example 2



Summary:

- High frequency error can be amplified by direct differentiation.
- Introduce a family of functions depending on a parameter h.
- A reasonable approximation can be obtained by a careful choice of h.
- How to choose an optimal h presents challenge.
- *A priori* information can improve the accuracy of approximation.
- Loss of information can not be avoided.

Example 2. (Inverse source problem)

Consider an object of unit mass moving along a straight line under a time dependent force $f(t)$. Assume that its initial position is at the origin and its initial velocity is 0. By Newton's second law (Force=mass $\times$ acceleration), the position $u(t)$ of the object at the instant t obeys

$$\begin{cases} \dfrac{d^2u}{dt^2} = f(t), \quad t \in [0,T] \\ u(0) = 0 \end{cases}$$

The inverse problem is to determine $f(t)$ from the measurement of $u(t)$ for all $t \in [0,T]$.

This is an ill-posed problem!

Integrating twice the equation gives

$$\frac{du}{dt}(t) = \int_0^t f(s)ds$$

and

$$u(t) = \int_0^t \int_0^\tau f(s)dsd\tau = \int_0^t \int_s^t f(s)d\tau ds.$$

We therefore obtain

$$u(t) = \int_0^t (t-s)f(s)ds$$

which is a Volterra integral equation of first kind.

Example 3. (X-ray tomography)

This is the inverse problem to determine f of a two-dimensional cross section Ω of human body by measuring the attenuation of X-rays. We assume f is supported in Ω, i.e. $f = 0$ outside Ω.

- X-rays travel along lines.
- We parametrize lines by their unit normal vector ω ($\|\omega\| = 1$) and their distance $s > 0$ to the origin, i.e $x = s\omega + t\omega^{\perp}$ represents the line, where $\omega^{\perp}$ denotes the unit normal vector orthogonal to ω.

We use I to denote the intensity of X-ray beams.

Assumption. the decay of the intensity of an X-ray beam along a distance Δt is proportional to the intensity I, the density f and Δt, i.e.

$$\Delta I(s\omega + t\omega^{\perp}) = -I(s\omega + t\omega^{\perp})f(s\omega + t\omega^{\perp})\Delta t.$$

Letting $\Delta t \to 0$ gives

$$\frac{dI}{dt}(s\omega + t\omega^{\perp}) = -I(s\omega + t\omega^{\perp})f(s\omega + t\omega^{\perp}).$$

Let $I_0(s,\omega)$ be the intensity of X-ray beam at the emitter and $I_1(s,\omega)$ the intensity at the detector. Then we have

$$\log I_1(s,\omega) - \log I_0(s,\omega) = \int_{\mathbb{R}} f(s\omega + t\omega^{\perp})dt.$$

Definition 1.2

We define

$$\mathcal{R}f(s,\omega) := \int_{\mathbb{R}} f(s\omega + t\omega^{\perp})dt$$

which is called the Radon transform of f.

Therefore, the inverse problem of determining the density f amounts to inverting Radon transform

$$\mathcal{R}f(s,\omega) = -\log \frac{I_1(s,\omega)}{I_0(s,\omega)}.$$

Special case: $\Omega := \{x \in \mathbb{R}^2 \ : \ |x| = 1\}$ and f is radial symmetric, i.e. $f(x) = F(|x|^2)$ for some single-variable function F.
It suffices to use X-rays parallel to the x-axis and consider only $\omega_0 = (0, \pm 1)$. Let $\tau = s^2 + t^2$. It can be shown that

$$\mathcal{R}f(s, \omega_0) = \int_{s^2}^{1} \frac{F(\tau)}{\sqrt{\tau - s^2}} d\tau$$

We introduce

$$g(s) = -\log \frac{I_1(s, \omega_0)}{I_0(s, \omega_0)}.$$

Then the determination of f can be obtained by solving the Abel integral equation

$$\int_{s^2}^{1} \frac{F(\tau)}{\sqrt{\tau - s^2}} d\tau = g(s).$$

It can be derived that explicit solution of the Abel equation

$$Af(t) := \int_{t}^{1} \frac{f(\tau)}{\sqrt{\tau - t}} d\tau = g(t).$$

is given by

$$f(t) = -\frac{1}{\pi}\frac{d}{dt}\left(\int_{1}^{t} \frac{g'(\tau)}{\sqrt{\tau - t}} d\tau\right).$$

Summary:

Many inverse problems can be reduced to solving the integral equation of the form

$$\int_\Omega K(x,y)f(y)dy = g(x), \quad x \in D$$

where Ω and D are domains in $\mathbb{R}^n$, $K(x,y)$ is a given function called kernel. This equation is called a *Fredholm equation of first kind*. We need to specify on which solution space the equation has to be solved.

2. Theory of Banach and Hilbert spaces

We will always assume that X is a linear space in which an addition and a scalar multiplication are defined with suitable properties, see any textbook on linear algebra for details.

Linear space has only an algebraic structure. In order to measure the length of elements and distance between any two elements, we need to introduce the norm.
Most of the proofs are omitted here, but they can be found in any linear algebra or analysis textbook.

Definition 2.1 (norm and normed spaces)

Let X be a complex (or real) linear space. A function $p : X \to \mathbb{R}$ is called a norm on X if it satisfies the following properties:

(a) Positivity: $p(x) \geq 0$ for all $x \in X$; $p(x) = 0$ if and only if $x = 0$.

(b) Homogeneity: $p(\alpha x) = |\alpha|\, p(x)$ for any $x \in X$ and $\alpha \in \mathbb{C}$ (or $\mathbb{R}$).

(c) Triangle inequality: $p(x + y) \leq p(x) + p(y)$.

We will write $\|x\| := p(x)$. A linear space X equipped with a norm is called a normed space.

In a normed space X, the distance between two elements $x, y \in X$ is defined as $\|x - y\|$.

Definition 2.2 (convergence)

Let X be a normed space. A sequence $\{x_n\}$ is called *convergent* if there is $x \in X$ such that $\lim_{n\to\infty} \|x_n - x\| = 0$. We will call x the *limit* of $\{x_n\}$ and write

$$\lim_{n\to\infty} x_n = x \quad \text{or} \quad x_n \to x \text{ as } n \to \infty$$

Lemma 2.3

The limit of a convergent sequence is uniquely determined.

Definition 2.4 (Cauchy sequence)

Let X be a normed space. A sequence $\{x_n\}$ is called a Cauchy sequence if

$$\lim_{m,n\to\infty} \|x_n - x_m\| = 0$$

Lemma 2.5

Any convergent sequence is a Cauchy sequence.

Definition 2.6 (completeness)

A normed space X is called complete if every Cauchy sequence in X is convergent. A complete normed space is called a Banach space.

Examples.

1) Given $1 \leq p < \infty$. Let ℓ^p denote the collection of all sequence $x = \{x_n\}$ satisfying

$$\sum_{n=1}^{\infty} |x_n|^p < \infty.$$

It is a linear space under the addition and scalar multiplication

$$\{x_n\} + \{y_n\} := \{x_n + y_n\}, \quad \alpha \{x_n\} := \{\alpha x_n\}.$$

For $x := \{x_n\}$ we define

$$\|x\| := \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

We can show that is is a norm on ℓ^p. Checking (a) and (b) in the definition of the norm is easy. Verifying (c) in the definition of norm amounts to proving the Minkowski inequality:

$$\left(\sum_{n=1}^{\infty} |x_n + y_n|^p \right)^{1/p} \leq \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p} + \left(\sum_{n=1}^{\infty} |y_n|^p \right)^{1/p}.$$

It can be shown that ℓ^p is a Banach space.

2) Let ℓ^∞ denote the collection of all sequence $x = \{x_n\}$ satisfying

$$\sup_n |x_n| < \infty$$

endowed with the same addition and scalar multiplication. Then

$$\|x\|_\infty := \sup_n |x_n|, \quad \forall x = (x_n) \in \ell^\infty$$

is a norm on ℓ^∞ and, under this norm, ℓ^∞ is a Banach space.

1) Let $C[0,1]$ denote the collection of all continuous function on $[0,1]$ with the usual addition of function and scalar multiplication. Then

$$\|f\| := \max_{t\in[0,1]} |f(t)|, \quad f \in C[0,1]$$

is a norm and $C[0,1]$ is a Banach space.

Examples.

1) In $C[0,1]$ we can define for each $1 \le p < \infty$

$$\|f\|_{L^p} := \left(\int_0^1 |f(t)|^p \, dt \right)^{1/p}, \quad f \in C[0,1].$$

Then $C[0,1]$ is a normed space but it is not a Banach space under this norm.

Any normed space can be completed to become a Banach space!

We use $L^p[0,1]$ to denote the completion of $C[0,1]$ under the above norm.

5) We use $L^\infty[0,1]$ to denote all measurable function such that it is a bounded function except on a measure zero set. It is a Banach space under the norm

$$\|f\|_{L^\infty} := \operatorname*{ess\,sup}_{t \in [0,1]} |f(t)|.$$

In order to measure the angle between any two elements in a linear space, it is necessary to introduce an inner product.

Definition 2.7 (inner product)

Let X be a complex (or real) linear space. A function $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{C}$ or $\mathbb{R}$ is called an inner product on X if it has the following properties:

(a) positivity: $\langle x, x \rangle \geq 0$ for all $x \in X$, and $\langle x, x \rangle = 0 \Leftrightarrow x = 0$

(b) symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all $x, y \in X$;

(c) linearity: for all $x, y, z \in X$ and $\alpha, \beta \in \mathbb{C}$ or $\mathbb{R}$ there holds

$$\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$$

where the bar denotes the complex conjugate.

Lemma 2.8

Let X *be a linear space with an inner product* $\langle \cdot, \cdot \rangle$*. Then*

$$\|x\| := \langle x, x \rangle^{1/2}, \quad \forall x \in X$$

defines a norm on X*.*

Lemma 2.9 (Cauchy-Schwarz inequality)

Let X *be a linear space with an inner product* $\langle \cdot, \cdot \rangle$*. Then for any* $x, y \in X$ *there holds*

$$|\langle x, y \rangle| \leq \|x\| \, \|y\| \, .$$

Definition 2.10 (Hilbert space)

A linear space with an inner product $\langle \cdot, \cdot \rangle$ is called a pre-Hilbert space. If it is complete with respect to the norm $\|x\| := \langle x, x \rangle^{1/2}$, then it is called a Hilbert space.

Lemma 2.11

In the pre-Hilbert space X there holds the parallelogram law:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad x, y \in X.$$

Lemma 2.12

Let X be a normed space with norm $\|\cdot\|$. Then X has an inner product $\langle \cdot, \cdot \rangle$ such that $\|x\| = \langle x, x \rangle^{1/2} \Leftrightarrow \|\cdot\|$ satisfies the parallelogram law.

Examples.

1) ℓ^2 is a Hilbert space with inner product

$$\langle x, y \rangle := \sum_{n=1}^{\infty} x_n \overline{y_n}$$

for any $x = \{x_n\}, y = \{y_n\} \in \ell^2$.

2) ℓ^p is a Banach space but not a Hilbert space for any $p \neq 2$. To see this, let

$$x = (1, 0, 0, \cdots), \quad y = (0, 1, 0, \cdots).$$

Then we have $\|x\| = \|y\| = 1$ and $\|x + y\| = \|x - y\| = 2^{1/p}$. Therefore

$$2^{1+2/p} = \|x + y\|^2 + \|x - y\|^2 \neq 2(\|x\|^2 + \|y\|^2) = 4$$

for $p \neq 2$. Thus the parallelogram law does not hold.

3) By a similar argument, we can see that ℓ^∞ is not a Hilbert space.

4) By a similar argument as above, we can see that $L^2[0,1]$ is a Hilbert space, while $L^p[0,1]$ is not for all $p \neq 2$.
5) $C[0,1]$ is not a Hilbert space. To see this, we take $f(t) = 1 + t$ and $g(t) = 2t$. Then

$$\|f\| = 2, \quad \|g\| = 2, \quad \|f+g\| = 4, \quad \|f-g\| = 1.$$

Therefore

$$17 = \|f+g\|^2 + \|f-g\|^2 \neq 2(\|f\|^2 + \|g\|^2) = 16$$

and thus the parallelogram law is violated.

Definition 2.13

Let $A : X \to Y$ be a map between normed spaces X and Y.

- A is called continuous at $x \in X$ if $\lim_{n\to\infty} \|Ax_n - Ax\| = 0$ for any sequence $\{x_n\} \subset X$ with $\lim_{n\to\infty} \|x_n - x\| = 0$.
- A is said to be continuous on X if it is continuous at every $x \in X$.
- A is called a linear operator if

$$A(\alpha x_1 + \beta x_2) = \alpha A x_1 + \beta A x_2$$

for all $x_1, x_2 \in X$ and $\alpha, \beta \in \mathbb{C}$ or $\mathbb{R}$.

- A is called a bounded linear operator if it is linear and there is a constant C such that

$$\|Ax\| \leq C \|x\|, \quad x \in X.$$

The proof of the following result can be read in any standard book in analysis.

Lemma 2.14

A linear operator is bounded if and only if it is continuous.

For a bounded linear operator $A : X \to Y$ between two Banach spaces X and Y, we define

$$\|A\| := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|$$

and call it the norm of A.

Definition 2.15

Any linear map $f : X \to \mathbb{C}$ or $\mathbb{R}$ on a linear space X is called a linear functional. We use $\|f\|$ to denote its norm, i.e.

$$\|f\| = \sup_{\|x\|=1} |f(x)| .$$

Let X be a normed space. We use X^* to denote the collection of all bounded linear functionals over X. It becomes a linear space under the addition and scalar multiplication

$$(f+g)(x) := f(x)+g(x), \quad (\alpha f)(x) = \alpha f(x), \quad \forall f, g \in X^*, \alpha \in \mathbb{C}$$

Moreover, $\|f\|$ defines a norm on X^* with which X^* is a Banach space. We will call X^* the dual space of X
. When X is a Hilbert space, we have Riesz representation theorem which says that for any $f \in X^*$ there is a unique $\xi \in X$ such that

$$\|f\| = \|\xi\| \quad \text{and} \quad f(x) = \langle x, \xi \rangle \text{ for all } x \in X.$$

Therefore X^* is isometric to X and we can identify X^* with X.

Let $A : X \to Y$ be a bounded linear operator between two Hilbert spaces X and Y, we define the map $A^* : Y \to X$ by

$$\langle A^* y, x \rangle = \langle y, Ax \rangle \,, \quad \forall x \in X, y \in Y$$

and call it the adjoint operator of A. Then we have

$$\begin{aligned} \|A^* y\| &= \sup_{\|x\|=1} |\langle A^* y, x \rangle| = \sup_{\|x\|=1} |\langle y, Ax \rangle| \\ &\le \sup_{\|x\|=1} \|Ax\| \, \|y\| = \|A\| \, \|y\| \,. \end{aligned}$$

Thus $A^* : Y \to X$ is a bounded linear operator with $\|A^*\| \le \|A\|$. We in fact have $\|A^*\| = \|A\|$.

- Let X be a Hilbert space. We call two elements $u, v \in X$ orthogonal if $\langle u, v \rangle = 0$ and write $u \perp v$.
- Let U be a subset of X we define

$$U^{\perp} := \{x \in X \ : \ \langle x, u \rangle = 0 \text{ for all } u \in U\}$$

  and call it the orthogonal complement of U in X.
- A sequence $\{u_n\}$ in X is called an orthogonal basis if

$$\langle u_n, u_m \rangle = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases}$$

  and any element $x \in X$ can be represented as

$$x = \sum_n \langle x, u_n \rangle \, u_n.$$

  Moreover, there holds the Parseval identity

$$\|x\|^2 = \sum_n |\langle x, u_n \rangle|^2 .$$

- There holds the Bessel's inequality. Suppose $\{u_1, u_2, \dots,\}$ is an orthonormal basis in X, then for any $x \in X$,

$$\sum_{n=1}^{\infty} |\langle x, u_n \rangle| \leq \|x\|^2 .$$

Let $A : X \to Y$ be a bounded linear operator between two Hilbert spaces. We define

$$\mathcal{N}(A) := \{x \in X : Ax = 0\}$$
$$\mathcal{R}(A) := \{y \in Y : y = Ax \text{ for some } x \in X\}$$

and call them the null space and range of A, respectively.

Lemma 2.16

$$\mathcal{N}(A)^\perp = \overline{\mathcal{R}(A^*)}, \quad \mathcal{N}(A^*)^\perp = \overline{\mathcal{R}(A)}$$
$$\mathcal{N}(A) = \overline{\mathcal{R}(A^*)}^\perp, \quad \mathcal{N}(A^*) = \overline{\mathcal{R}(A)}^\perp$$

where the bar denotes closure.

Definition 2.17

A linear operator $A : X \to Y$ is called compact if for any bounded sequence $\{x_n\}$ in X, i.e. $\|x_n\| \leq M$ for some number M, the sequence $\{Ax_n\}$ contains a convergent subsequence in Y.

Remark 2.1

- Compact linear operators are bounded.
  If not, we can find $\{x_n\}$ with $\|x_n\| = 1$ such that $\|Ax_n\| \to \infty$ as $n \to \infty$. Since $\|x_n\|$ is bounded, by compactness, $\|Ax_n\|$ has a convergent subsequence, contradiction.
- The identity map $I : X \to X$ is compact if and only if $\dim X < \infty$.

Theorem 2.18 (singular value decomposition)

Let $A : X \to Y$ be a compact linear operator between two Hilbert spaces X and Y. Then there exists a countable set of triples (σ_n, u_n, v_n) with the following properties:

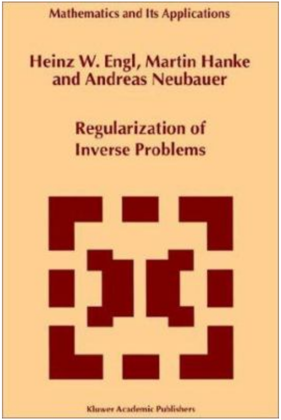
- $\sigma_1 \geq \sigma_2 \geq \cdots > 0$ *with* $\sigma_1 = \|A\|$*; and, if* $\dim \mathcal{R}(A) = \infty$*, then* $\lim_{n\to\infty} \sigma_n = 0$*;*
- $\{u_n\}$ *is an orthonormal basis of* $\mathcal{N}(A)^{\perp}$*;*
- $\{v_n\}$ *is an orthonormal basis of* $\mathcal{N}(A^*)^{\perp}$*;*
- $Au_n = \sigma_n v_n$ *and* $A^* v_n = \sigma_n u_n$*.*

We call σ_n the singular values, u_n the right singular vectors, and v_n the left singular vectors of A, respectively. We also call (σ_n, u_n, v_n) the singular system of A.

Remark 2.2

- Observe that
$$A^*Au_n = \sigma_n A^*v_n = \sigma_n^2 u_n.$$
Therefore σ_n are the square roots of the eigenvalues of A^*A repeated according to their multiplicity, i.e. according to the dimension of the null spaces $\mathcal{N}(\sigma_n^2 I - A^*A)$, and u_n are the corresponding eigenvectors.
- If $A : X \to X$ is self-adjoint and positive definite, then the singular values σ_n of A equal its eigenvalues, the right singular vectors equal the eigenvectors, and the left singular vector equal $\frac{1}{\sigma_n}Au_n$.
- If the singular system of A is (σ_n, u_n, v_n), then (σ_n, v_n, u_n) is the singular system of A^*.

Suggested reference: *Regularization of Inverse Problems* by H. Engl, M. Hanke, and A. Neubauer (2000), aka the bible of regularization theory.



Let $A : X \to Y$ be a compact linear operator between two Hilbert spaces X and Y with singular system $\{(\sigma_n, u_n.v_n)\}$. Then for any $x \in X$ we have

$$x = x_0 + x^\perp \text{ with } x_0 \in \mathcal{N}(A) \text{ and } x^\perp = \sum_n \langle x, u_n \rangle u_n \in \mathcal{N}(A)^\perp,$$

$$Ax = \sum_n \sigma_n \langle x, u_n \rangle v_n, \quad A^*Ax = \sum_n \sigma_n^2 \langle x, u_n \rangle u_n.$$

An induction argument shows for all integer $k \geq 1$ that

$$(A^*A)^k x = \sum_n \sigma_n^{2k} \langle x, u_n \rangle u_n$$

This implies that for any polynomial $p(\lambda) = \sum_{j=0}^N a_j \lambda^j$ there holds

$$
\begin{aligned}
p(A^*A) &= a_0 x + \sum_{j=1}^{N} a_j \sum_n \sigma_n^{2j} \langle x, u_n \rangle u_n \\
&= a_0(x - x^\perp) + \sum_n p(\sigma_n^2) \langle x, u_n \rangle u_n \\
&= p(0)(x - x^\perp) + \sum_n p(\sigma_n^2) \langle x, u_n \rangle u_n.
\end{aligned}
$$

This motivates the following definition.

Definition 2.19

Let f be a continuous function defined on $[0, \|A\|^2]$. We define $f(A^*A) : X \to X$ by

$$
f(A^*A)x := f(0)(x - x^\perp) + \sum_n f(\sigma_n^2) \langle x, u_n \rangle u_n.
$$

It also follows that $f(A^*A) : X \to X$ is a bounded linear operator with

$$\|f(A^*A)\| \leq \sup_{\lambda \in [0, \|A\|^2]} |f(\lambda)|\,.$$

Lemma 2.20

$$f(AA^*)A = Af(A^*A), \quad f(A^*A)A^* = A^*f(AA^*)$$

Lemma 2.21 (interpolation inequality)

For all $q > r \geq 0$, and for all $x \in X$, there holds

$$\|(A^*A)^r x\| \leq \|(A^*A)^q x\|^{\frac{r}{q}} \, \|x\|^{1-\frac{r}{q}}$$

3. Regularization methods in Hilbert spaces

Let X and Y be two Hilbert spaces and let $A : X \to Y$ be a bounded linear operator. We consider the operator equation

$$Ax = y.$$

Definition 3.1

The above equation is called well-posed in the sense of Hadamard if

- for any $y \in Y$ the equation has a solution in X;
- the equation has a unique solution in X;
- the solution depends continuously on the data, i.e. for any sequence $\{y_n\}$ that converges to y in Y , let $\{x_n\}$ be the solution of $Ax = y_n$, then $\{x_n\}$ converges to the solution of the equation $Ax = y$.

If one of the requirements is violated, the equation is called ill-posed.

Theorem 3.2 (Picard)

Let $A : X \to Y$ be a compact linear operator between two Hilbert spaces X and Y with singular system $\{(\sigma_n, u_n, v_n)\}$. Then the equation

$$Ax = y$$

is solvable if and only if

$$y \in \mathcal{N}(A^*)^\perp \quad \textit{and} \quad \sum_{n=1}^{\infty} \frac{1}{\sigma_n^2} \left|\langle y, v_n \rangle\right|^2 < \infty.$$

In this case,

$$x^\dagger := \sum_{n=1}^{\infty} \frac{1}{\sigma_n} \langle y, v_n \rangle\, u_n$$

is the unique solution with minimal norm.

Proof. Assume first that $Ax = y$ has a solution x_*. Then for any $v \in \mathcal{N}(A^*)$ we have

$$\langle y, v\rangle = \langle Ax_*, v\rangle = \langle x_*, A^*v\rangle = \langle x_*, 0\rangle = 0$$

which implies $y \in \mathcal{N}(A^*)^\perp$. Moreover,

$$\langle y, v_n\rangle = \langle Ax_*, v_n\rangle = \langle x_*, A^*v_n\rangle = \sigma_n \langle x_*, u_n\rangle = 0$$

which, by Bessel's inequality,

$$\sum_{n=1}^{\infty} \frac{1}{\sigma_n^2} \left|\langle x_*, u_n\rangle\right|^2 \leq \left\|x_*\right\|^2 < \infty.$$

Conversely, by the summability condition we have

$$\left\|\sum_{n=m}^{k}\frac{1}{\sigma_n}\langle y, v_n\rangle u_n\right\|^2 = \sum_{n=m}^{k}\frac{1}{\sigma_n^2}\langle y, v_n\rangle^2 \to 0 \text{ as } m, k \to \infty.$$

This implies that the series defining $x^\dagger$ is convergent in X and thus $x^\dagger$ is well-defined with $x^\dagger \in \mathcal{N}(A)^\perp$. Because $y \in \mathcal{N}(A^*)^\perp$,

$$Ax^\dagger = \sum_{n=1}^{\infty}\frac{1}{\sigma_n}\langle y, v_n\rangle Au_n = \sum_{n=1}^{\infty}\langle y, v_n\rangle v_n = y.$$

Finally let x_* be any solution of $Ax = y$. Then $A(x_* - x^\dagger) = 0$ which implies that $x_* - x^\dagger \in \mathcal{N}(A)$. Since $x^\dagger \in \mathcal{N}(A)^\perp$, we have

$$\|x_*\|^2 = \left\|x_* - x^\dagger + x^\dagger\right\|^2 = \left\|x_* - x^\dagger\right\|^2 + \left\|x^\dagger\right\|^2.$$

This implies that $\|x_*\| \geq \left\|x^\dagger\right\|$ and $\|x_*\| = \left\|x^\dagger\right\| \Leftrightarrow x_* = x^\dagger$. $\square$

Remark 3.1

Assume that $Ax = y$ is solvable with $x^\dagger$ given in Theorem 3.2. Consider the perturbation of y given by

$$y_k = y + \sqrt{\sigma_k} v_k$$

Then we have $\|y_k - y\| = \sqrt{\sigma_k}$. Moreover, the equation $Ax = y_k$ is solvable with minimal norm solution given by

$$x_k^\dagger := \sum_{n=1}^{\infty} \frac{1}{\sigma_n} \langle y_k, v_n \rangle u_n = \sum_{n=1}^{\infty} \frac{1}{\sigma_n} \langle y, v_n \rangle u_n + \frac{1}{\sqrt{\sigma_k}} u_k = x^\dagger + \frac{1}{\sqrt{\sigma_k}} u_k$$

So $\left\| x_k^\dagger - x^\dagger \right\| = \frac{1}{\sqrt{\sigma_k}}$. If $\dim \mathcal{R}(A) = \infty$, then $\lim_{k\to\infty} \sigma_k = 0$. Thus

$$\lim_{k\to\infty} \|y_k - y\| = 0 \quad \text{text} \quad \lim_{k\to\infty} \left\| x_k^\dagger - x^\dagger \right\| = \infty.$$

This shows that the problem is ill-posed.

3.1. Tikhonov regularization

We consider again the equation

$$Ax = y.$$

By using noisy data y^δ satisfying $\left\|y^\delta - y\right\|$, we will produce approximate solution to the minimal norm solution $x^\dagger$ of $Ax = y$. A natural idea is to consider the optimization problem:

$$\text{minimize } \|x\| \quad \text{subject to } \left\|Ax - y^\delta\right\| \leq \delta.$$

By the method of Lagrange multiplier we may consider minimizing the function $\|x\|^2 + \lambda(\left\|Ax - y^\delta\right\|^2 - \delta^2)$ over$= X$, or equivalently

$$\text{minimize } \left\|Ax - y^\delta\right\|^2 + \alpha \|x\|^2 \quad \text{over } X.$$

Lemma 3.3

For each $\alpha > 0$ the minimization problem

$$\min_{x \in X} \left\{ \left\| Ax - y^{\delta} \right\|^2 + \alpha \left\| x \right\|^2 \right\}$$

has a unique minimizer.

The proof can be done using two stages: existence and uniqueness.

We need some preparation.

Definition 3.4 (weak convergence)

A sequence $\{x_n\}$ in a Hilbert space X is said to weakly converge to an element $x \in X$ if

$$\lim_{n\to\infty} \langle z, x_n \rangle = \langle z, x \rangle, \quad \forall z \in X.$$

We will write $x_n \rightharpoonup x$ for weak convergence.

Lemma 3.5

(a) *Any weakly convergent sequence in Hilbert spaces is bounded.*

(b) *Any bounded sequence in Hilbert spaces contains a weakly convergent subsequence.*

Proof. We omit the proof.

Definition 3.6 (weak lower semi-continuity)

A function f defined in a Hilbert space X is said to be weakly lower semi-continuous if for any sequence $\{x_n\}$ that converges weakly to $x \in X$ there holds

$$f(x) \leq \liminf_{n\to\infty} f(x_n).$$

Lemma 3.7

The norm in Hilbert spaces is weakly lower semi-continuous, i.e. if $x_n \rightharpoonup x$ as $n \to \infty$ then $\|x\| \leq \liminf_{n\to\infty} \|x_n\|$.

Proof. Since $x_n \rightharpoonup x$, from Cauchy-Schwarz inequality we have

$$\|x\|^2 = \langle x, x\rangle = \lim_{n\to\infty} \langle x, x_n\rangle = \liminf_{n\to\infty} \langle x, x_n\rangle \leq \liminf_{n\to\infty} \|x\| \, \|x_n\| \, .$$

This implies the result. □

Lemma 3.8

For each $\alpha > 0$,the minimization problem

$$\min_{x \in X} \left\{ \left\| Ax - y^\delta \right\|^2 + \alpha \left\| x \right\|^2 \right\}$$

has a unique minimizer.

Proof. Existence: For each $\alpha > 0$ we set

$$m_\alpha := \inf_{x \in X} \left\{ \left\| Ax - y^\delta \right\|^2 + \alpha \left\| x \right\|^2 \right\}$$

Then $0 \leq m_\alpha \leq \left\| y^\delta \right\|^2 < \infty$. Let $\{x_n\}$ be a minimizing sequence, i.e.

$$\left\| Ax_n - y^\delta \right\|^2 + \alpha \left\| x_n \right\|^2 \to m_\alpha \text{ as } n \to \infty.$$

This implies that $\{x_n\}$ is a bounded sequence in X. Thus $\{x_n\}$ has a subsequence, still denoted it by $\{x_n\}$, that converges weakly to some $\hat{x} \in X$. Since A is bounded, we have for any $v \in Y$ that

$$\langle Ax_n, v\rangle = \langle x_n, A^* v\rangle \to \langle \hat{x}, A^* v\rangle\,, \quad \text{as } n \to \infty.$$

Thus $Ax_n \rightharpoonup A\hat{x}$ as $n \to \infty$. By the weak lower semi-continuity of norms in Hilbert spaces, we have

$$\|\hat{x}\| \le \liminf_{n\to\infty} \|x_n\|\,, \quad \left\|A\hat{x} - y^\delta\right\| \le \liminf_{n\to\infty} \left\|Ax_n - y^\delta\right\|.$$

Consequently

$$\begin{aligned} m_\alpha &\le \left\|A\hat{x} - y^\delta\right\|^2 + \alpha\,\|\hat{x}\|^2 \le \liminf_{n\to\infty} \left\|Ax_n - y^\delta\right\|^2 + \alpha \liminf_{n\to\infty} \|x_n\|^2 \\ &\le \liminf_{n\to\infty} \left(\left\|Ax_n - y^\delta\right\|^2 + \alpha\,\|x_n\|^2\right) = m_\alpha \end{aligned}$$

Therefore $m_\alpha = \left\|A\hat{x} - y^\delta\right\|^2 + \alpha\,\|\hat{x}\|^2$, i.e. $\hat{x}$ is a minimizer.

Uniqueness: Assume there exist two minimizers x_0 and x_1. For $\lambda \in (0,1)$ let $x_\lambda := (1-\lambda)x_0 + \lambda x_1$. Then

$$\left\|Ax_\lambda - y^\delta\right\|^2 + \alpha \left\|x_\lambda\right\|^2 \geq \left\|Ax_i - y^\delta\right\|^2 + \alpha \left\|x_i\right\|^2, \quad i = 0, 1.$$

Let $T_\alpha(x) := \left\|Ax - y^\delta\right\|^2 + \alpha \left\|x\right\|^2$. Then $T_\alpha(x_\lambda) \geq T_\alpha(x_i)$ for $i = 0, 1$. Use the identity $T_\alpha(x_\lambda) = (1-\lambda)T_\alpha(x_\lambda) + \lambda T_\alpha(x_\lambda)$. This implies

$$\begin{aligned} \left\|Ax_\lambda - y^\delta\right\|^2 + \alpha \left\|x_\lambda\right\|^2 \geq & (1-\lambda)\left\{\left\|Ax_0 - y^\delta\right\|^2 + \alpha \left\|x_0\right\|^2\right\} \\ & + \lambda \left\{\left\|Ax_1 - y^\delta\right\|^2 + \alpha \left\|x_1\right\|^2\right\} \end{aligned}$$

On the other hand, by substituting x_λ and observing that $-y^\delta = -(1-\lambda)y^\delta - \lambda y^\delta$, and by inner products, we get

$$\begin{aligned}\left\|Ax_\lambda - y^\delta\right\|^2 =& \left\|(1-\lambda)(Ax_0 - y^\delta) + \lambda(Ax_1 - y^\delta)\right\|^2 \\ =&(1-\lambda)^2 \left\|Ax_0 - y^\delta\right\|^2 + \lambda^2 \left\|Ax_1 - y^\delta\right\|^2 \\ &+ 2\lambda(1-\lambda)\left\langle Ax_0 - y^\delta, Ax_1 - y^\delta\right\rangle\end{aligned}$$

and

$$\begin{aligned}\|x_\lambda\|^2 &= \|(1-\lambda)x_0 + \lambda x_1\|^2 \\ &= (1-\lambda)^2\|x_0\|^2 + \lambda^2\|x_1\|^2 + 2\lambda(1-\lambda)\langle x_0, x_1\rangle.\end{aligned}$$

Note we used the fact for any $x, y \in X$,

$$\|x \pm y\|^2 = \|x\|^2 \pm 2\langle x, y\rangle + \|y\|^2.$$

Therefore

$$
\begin{aligned}
&\|Ax_\lambda - y^\delta\|^2 + \alpha \|x_\lambda\|^2 \\
=&(1-\lambda)^2 \left\{ \left\|Ax_0 - y^\delta\right\|^2 + \alpha \|x_0\|^2 \right\} \\
&+ \lambda^2 \left\{ \left\|Ax_1 - y^\delta\right\|^2 + \alpha \|x_1\|^2 \right\} \\
&+ 2\lambda(1-\lambda) \left\{ \left\langle Ax_0 - y^\delta, Ax_1 - y^\delta \right\rangle + \alpha \langle x_0, x_1 \rangle \right\}.
\end{aligned}
$$

This implies that

$$
\begin{aligned}
-\lambda(1-\lambda) \left\{ \left\|Ax_0 - y^\delta\right\|^2 + \alpha \|x_0\|^2 + \left\|Ax_1 - y^\delta\right\|^2 + \alpha \|x_1\|^2 \right\} \\
+ 2\lambda(1-\lambda) \left\{ \left\langle Ax_0 - y^\delta, Ax_1 - y^\delta \right\rangle + \right\} \geq 0
\end{aligned}
$$

Regrouping terms and by inner products, this implies

$$-\lambda(1-\lambda)\left\{\|Ax_0 - Ax_1\|^2 + \alpha\|x_0 - x_1\|^2\right\} \geq 0.$$

Therefore $x_0 = x_1$ and uniqueness follows. $\square$

We will denote the unique minimizer by x_α^δ, call it the Tikhonov regularized solution, and write

$$x_\alpha^\delta := \arg\min_{x\in X}\left\{\left\|Ax - y^\delta\right\|^2 + \alpha\|x\|^2\right\}.$$

Lemma 3.9

For each $\alpha > 0$, the minimizer x_α^δ satisfies the first order optimality equation

$$\alpha x_\alpha^\delta + A^*(Ax_\alpha^\delta - y^\delta) = 0.$$

Proof. For any $\varphi \in X$ consider the function

$$f(t) = \left\| A(x_\alpha^\delta + t\varphi) \right\|^2 + \alpha \left\| x_\alpha^\delta + t\varphi \right\|^2 .$$

Since x_α^δ is the minimizer, f achieves its minimum at $t = 0$ and thus $f'(t) = 0$. Observing that

$$\begin{aligned} f(t) = {} & \left\| Ax_\alpha^\delta - y^\delta \right\|^2 + 2t \left\langle Ax_\alpha^\delta - y^\delta, A\varphi \right\rangle + t^2 \left\| A\varphi \right\|^2 \\ & + \alpha \left(\left\| x_\alpha^\delta \right\|^2 + 2t \left\langle x_\alpha^\delta, \varphi \right\rangle + t^2 \left\| \varphi \right\|^2 \right) . \end{aligned}$$

Therefore

$$0 = f'(0) = 2 \left\langle Ax_\alpha^\delta - y^\delta, A\varphi \right\rangle + 2\alpha \left\langle x_\alpha^\delta, \varphi \right\rangle = 2 \left\langle A^*(Ax_\alpha^\delta - y^\delta) + \alpha x_\alpha^\delta,$$

Since $\varphi \in X$ is arbitrary, we have $A^*(Ax_\alpha^\delta - y^\delta) + \alpha x_\alpha^\delta = 0$.

In order to write x_α^δ explicitly, we need the invertibility of the operator $\alpha I + A^*A : X \to X$. Observing that

$$\langle (\alpha I + A^*A)x, x\rangle = \alpha \langle x, x\rangle + \langle Ax, Ax\rangle \geq \alpha \|x\|^2 ,$$

i.e. $\alpha I + A^*A$ is coercive, by the Lax-Milgram theorem we can conclude that $\alpha I + A^*A$ is invertible and write its inverse as $(\alpha I + A^*A)^{-1}$.

Recall that $\alpha x_\alpha^\delta + A^*(Ax_\alpha^\delta - y^\delta) = 0.$. This gives

$$(\alpha I + A^*A)x_\alpha^\delta = A^*y^\delta.$$

Therefore

$$x_\alpha^\delta = (\alpha I + A^*A)^{-1}A^*y^\delta.$$

We will approximate $x^\dagger$ by x_α^δ with suitable choice of α.

Numerical simulation. We consider the integral equation

$$(Ax)(s) := \int_0^1 K(s,t)x(t)dt = y(s)$$

where the kernel K is given by

$$K(s,t) = \begin{cases} 40s(1-t), & s \leq t, \\ 40t(1-s), & s \geq t. \end{cases}$$

The Tikhonov regularization takes the form

$$\min_{x \in L^2[0,1]} \left\{ \int_0^1 \left| Ax(s) - y^\delta(s) \right|^2 ds + \alpha \int_0^1 |x(s)|^2 ds \right\}$$

After discretization as before, the regularized solution satisfies

$$\left(\alpha \mathbf{I} + \mathbf{A}^T \mathbf{A}\right) \mathbf{x}_\alpha = \mathbf{A}^T \mathbf{y}^\delta.$$

First simulation: The true solution is given by $x^\dagger(t) = \sin(4\pi t)$. We use y^δ satisfying $\left\|y^\delta - y\right\|_{L^2[0,1]} = \delta$ with $\delta = 0.01$.
Second simulation: The true solution is given solution is

$$x^\dagger(t) = \begin{cases} 1, & \text{on } [0.25, 0.30] \text{ and } [0.70, 0.75] \\ -1, & \text{on } [0.475, 0.525] \\ 0, & \text{ elsewhere} \end{cases}$$

We use y^δ satisfying $\left\|y^\delta - y\right\|_{L^2[0,1]}$ with $\delta = 5.0 \times 10^{-3}$.

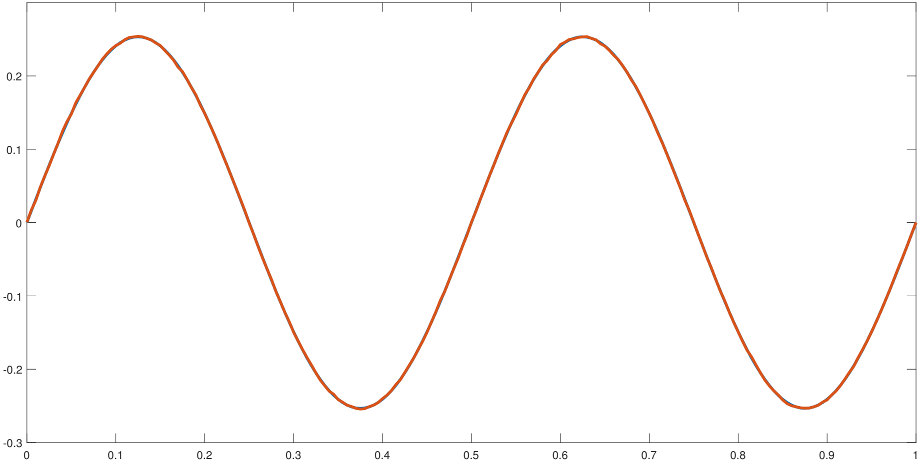
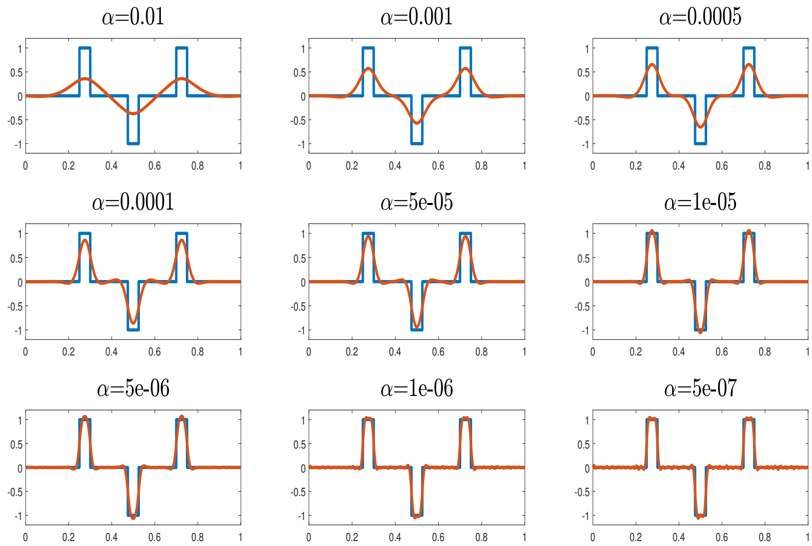
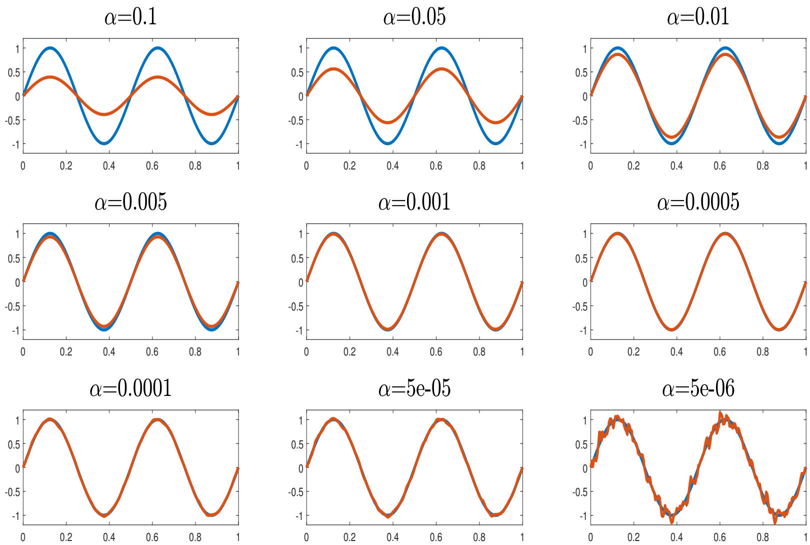


Figure: Exact data and noisy data with $\delta = 0.01$

Example 7



Example 8


α=0.1
α=0.05
α=0.01
α=0.005
α=0.001
α=0.0005
α=0.0001
α=5e-05
α=5e-06
1
0.5
0
-0.5
-1
0
0.2
0.4
0.6
0.8
1

Morozov's discrepancy principle. Let $\tau > 1$ be a given number.

- If $\|y^\delta\| \le \tau\delta$. we take $\alpha = \infty$, i.e. we use $x_\alpha^\delta := 0$ to approximate $x^\dagger$.
- If $\|y^\delta\| > \tau\delta$, we take $\alpha = \alpha(\delta)$ as the root of the equation

$$\|Ax_\alpha^\delta - y^\delta\| = \tau\delta$$

  and use $x_{\alpha(\delta)}^\delta$ to approximate $x^\dagger$.

It can be shown that this principle is well-defined. We only need to consider the case that $\|y^\delta\| > \tau\delta$. It suffices to show that

- $\alpha \mapsto \|Ax_\alpha^\delta - y^\delta\|$ is continuous on $(0, \infty)$;
- $\alpha \mapsto \|Ax_\alpha^\delta - y^\delta\|$ is strictly on $(0, \infty)$;
- $\lim_{\alpha\to\infty} \|Ax_\alpha^\delta - y^\delta\| = \|y^\delta\|$ and $\lim_{\alpha\to 0} \|Ax_\alpha^\delta - y^\delta\| = \|y^\delta\|$

Lemma 3.10

$$\lim_{\alpha\to\infty} \left\|Ax_\alpha^\delta - y^\delta\right\| = \left\|y^\delta\right\| \textit{ and } \lim_{\alpha\to 0} \left\|Ax_\alpha^\delta - y^\delta\right\| \leq \delta.$$

Proof. By the minimizing property of x_α^δ we have

$$\alpha \left\|x_\alpha^\delta\right\|^2 \leq \left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \alpha \left\|x_\alpha^\delta\right\|^2 \leq \left\|y^\delta\right\|^2.$$

This implies $\lim_{\alpha\to\infty} \left\|x_\alpha^\delta\right\| = 0$, and thus as $\alpha \to \infty$, $x_\alpha^\delta = 0$ and hence $\lim_{\alpha\to\infty} \left\|Ax_\alpha^\delta - y^\delta\right\| = \left\|y^\delta\right\|$.
By using the minimizing property of x_α^δ again we have

$$\left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \alpha \left\|x_\alpha^\delta\right\|^2 \leq \left\|Ax^\dagger - y^\delta\right\|^2 + \alpha \left\|x^\dagger\right\|^2 \leq \delta^2 + \alpha \left\|x^\dagger\right\|^2.$$

This implies $\lim_{\alpha\to 0} \left\|Ax_\alpha^\delta - y^\delta\right\|^2 \leq \delta^2$. □

Lemma 3.11

The function $\alpha \mapsto x_\alpha^\delta$ is continuous on X for $\alpha \in (0, \infty)$.

Proof. By the minimizing property of x_α^δ we have

$$\left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \alpha \left\|x_\alpha^\delta\right\|^2 \leq \left\|Ax_\beta^\delta - y^\delta\right\|^2 + \alpha \left\|x_\beta^\delta\right\|^2 .$$

We consider what happens when α gets close to β. Regrouping terms, and adding the same terms both sides, we get

$$\begin{aligned} &\|A(x_\alpha^\delta - x_\beta^\delta)\|^2 + \alpha \left\|x_\alpha^\delta - x_\beta^\delta\right\|^2 \\ &\quad \leq \left\|Ax_\beta^\delta - y^\delta\right\|^2 - \left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \|A(x_\alpha^\delta - x_\beta^\delta)\|^2 \\ &\qquad + \alpha \left(\left\|x_\alpha^\delta\right\|^2 - \left\|x_\beta^\delta\right\|^2 + \left\|x_\alpha^\delta - x_\beta^\delta\right\|^2\right) \\ &\quad = -2 \left\langle A(x_\alpha^\delta - x_\beta^\delta), Ax_\beta^\delta - y^\delta \right\rangle - 2\alpha \left\langle x_\alpha^\delta - x_\beta^\delta, x_\beta^\delta \right\rangle \\ &\quad = -2 \left\langle x_\alpha^\delta - x_\beta^\delta, A^*(Ax_\beta^\delta - y^\delta) \right\rangle - 2\alpha \left\langle x_\alpha^\delta - x_\beta^\delta, x_\beta^\delta \right\rangle . \end{aligned}$$

By using the first order optimality equation for x_β^δ, $A^*(Ax_\beta^\delta - y^\delta) = -\beta x_\beta^\delta$. Then by Cauchy-Schwarz inequality we have

$$\alpha \left\| x_\alpha^\delta - x_\beta^\delta \right\|^2 \leq 2(\beta-\alpha) \left\langle x_\alpha^\delta - x_\beta^\delta, x_\beta^\delta \right\rangle \leq 2 \left| \beta - \alpha \right| \left\| x_\alpha^\delta - x_\beta^\delta \right\| \left\| x_\beta^\delta \right\|.$$

This implies

$$\left\| x_\alpha^\delta - x_\beta^\delta \right\| \leq \frac{2 \left| \beta - \alpha \right|}{\alpha} \left\| x_\beta^\delta \right\|.$$

By the minimizing property of x_β^δ we have $\beta \left\| x_\beta^\delta \right\|^2 \leq \left\| y^\delta \right\|^2$. Therefore

$$\left\| x_\alpha^\delta - x_\beta^\delta \right\| \leq \frac{2 \left\| y^\delta \right\|}{\alpha \sqrt{\beta}} \left| \alpha - \beta \right|.$$

Thus if $\alpha \to \beta$, $x_\alpha^\delta \to x_\beta^\delta$. This shows the continuity. $\square$

Lemma 3.12

For any $0 < \alpha < \beta$ *there hold*

$$\left\|x_\alpha^\delta\right\| \geq \left\|x_\beta^\delta\right\| \text{ and } \left\|Ax_\alpha^\delta - y^\delta\right\| \leq \left\|Ax_\beta^\delta - y^\delta\right\|.$$

Proof. From the minimizing properties of x_α^δ and x_β^δ it follows that

$$\begin{aligned}
\left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \alpha\left\|x_\alpha^\delta\right\|^2 &\leq \left\|Ax_\beta^\delta - y^\delta\right\|^2 + \alpha\left\|x_\beta^\delta\right\|^2 \\
\left\|Ax_\beta^\delta - y^\delta\right\|^2 + \beta\left\|x_\beta^\delta\right\|^2 &\leq \left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \beta\left\|x_\alpha^\delta\right\|^2.
\end{aligned}$$

Combining these two equations we obtain

$$\begin{aligned}
\left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \alpha\left\|x_\alpha^\delta\right\|^2 &\leq \left\|Ax_\beta^\delta - y^\delta\right\|^2 + \beta\left\|x_\beta^\delta\right\|^2 + (\alpha - \beta)\left\|x_\beta^\delta\right\|^2 \\
&\leq \left\|Ax_\alpha^\delta - y^\delta\right\|^2 + \beta\left\|x_\alpha^\delta\right\|^2 + (\alpha - \beta)\left\|x_\beta^\delta\right\|^2
\end{aligned}$$

Therefore

$$(\alpha - \beta)\left(\left\|x_\alpha^\delta\right\|^2 - \left\|x_\beta^\delta\right\|^2\right) \le 0$$

Since $\alpha - \beta < 0$, we can get the first inequality. Using this and the minimizing property of x_α^δ we further obtain

$$\left\|Ax_\alpha^\delta - y^\delta\right\|^2 - \left\|Ax_\beta^\delta - y^\delta\right\|^2 \le \alpha\left(\left\|x_\beta^\delta\right\|^2 - \left\|x_\alpha^\delta\right\|^2\right) \le 0$$

which gives the second inequality. □

Lemma 3.13

When $\left\|y^\delta\right\| > \delta$*, the function*

$$\alpha \mapsto \left\|Ax_\alpha^\delta - y^\delta\right\|$$

is strictly increasing on $(0, \infty)$*.*

Proof. If the result is not true, then there exist $0 < \alpha_1 < \alpha_2$ such that

$$\left\| Ax^\delta_{\alpha_1} - y^\delta \right\| = \left\| Ax^\delta_{\alpha_2} - y^\delta \right\|.$$

By using the minimizing property of $x^\delta_{\alpha_1}$ we have

$$\begin{aligned} \left\| Ax^\delta_{\alpha_1} - y^\delta \right\|^2 + \alpha_1 \left\| x^\delta_{\alpha_1} \right\|^2 &\leq \left\| Ax^\delta_{\alpha_2} - y^\delta \right\|^2 + \alpha_1 \left\| x^\delta_{\alpha_2} \right\|^2 \\ &= \left\| Ax^\delta_{\alpha_1} - y^\delta \right\|^2 + \alpha_1 \left\| x^\delta_{\alpha_2} \right\|^2 \end{aligned}$$

So $\left\| x^\delta_{\alpha_1} \right\| \leq \left\| x^\delta_{\alpha_2} \right\|$. By Lemma 3.12 we also have $\left\| x^\delta_{\alpha_2} \right\| \leq \left\| x^\delta_{\alpha_1} \right\|$. Thus $\left\| x^\delta_{\alpha_1} \right\| = \left\| x^\delta_{\alpha_2} \right\|$. Consequently

$$x^\delta_{\alpha_1} = \arg\min_{x \in X} \left\{ \left\| Ax - y^\delta \right\|^2 + \alpha_2 \left\| x \right\|^2 \right\}$$

By uniqueness, $x^\delta_{\alpha_1} = x^\delta_{\alpha_2}$. By the first order optimality equation on minimizers, we have

$$\alpha_1 x^\delta_{\alpha_1} + A^* \left(A x^\delta_{\alpha_1} - y^\delta \right) = 0,$$
$$\alpha_2 x^\delta_{\alpha_1} + A^* \left(A x^\delta_{\alpha_1} - y^\delta \right) = 0,$$

Therefore $(\alpha_1 - \alpha_2) x^\delta_{\alpha_1} = 0$. This implies $x^\delta_{\alpha_1} = 0$ and thus $A^* y^\delta = 0$. So by assumption on y^δ,

$$\begin{aligned} 0 &= \left\langle A^* y^\delta, x^\dagger \right\rangle = \left\langle y^\delta, A x^\dagger \right\rangle = \left\langle y^\delta, y \right\rangle = \left\langle y^\delta, y^\delta \right\rangle + \left\langle y^\delta, y - y^\delta \right\rangle \\ &\geq \left\| y^\delta \right\|^2 - \left\| y^\delta \right\| \left\| y^\delta - y \right\| \geq \left\| y^\delta \right\| \left(\left\| y^\delta \right\| - \delta \right) > 0 \end{aligned}$$

We therefore obtain a contradiction. □

We thus showed that the discrepancy principle is well-defined.

Theorem 3.14

Let $\alpha(\delta)$ be determined by Morozov's discrepancy principle with $\tau > 1$. Then

$$\lim_{\delta \to 0} \left\| x^\delta_{\alpha(\delta)} - x^\dagger \right\| = 0.$$

Proof. If $y = 0$, then $x^\dagger = 0$ and $\left\| y^\delta \right\| \leq \delta$. This implies $\alpha(\delta) = \infty$ and thus $x^\delta_{\alpha(\delta)} = 0 = x^\dagger$. The assertion is trivial.

We may assume $y \neq 0$. Then $\left\| y^\delta \right\| \geq \|y\| - \delta > \tau\delta$ for sufficiently small $\delta > 0$. Thus $0 < \alpha(\delta) < \infty$ and

$$\left\| A x^\delta_{\alpha(\delta)} - y^\delta \right\| = \tau\delta.$$

By the minimizing property of $x^\delta_{\alpha(\delta)}$ we have

$$\begin{aligned} \tau^2\delta^2 + \alpha(\delta) \left\| x^\delta_{\alpha(\delta)} \right\|^2 &= \left\| A x^\delta_{\alpha(\delta)} - y^\delta \right\|^2 + \alpha(\delta) \left\| x^\delta_{\alpha(\delta)} \right\|^2 \\ &\leq \left\| A x^\dagger - y^\delta \right\|^2 + \alpha(\delta) \left\| x^\dagger \right\|^2 \\ &\leq \delta^2 + \alpha(\delta) \left\| x^\dagger \right\|^2. \end{aligned}$$

Since $\tau > 1$, this implies

$$\left\|x^\delta_{\alpha(\delta)}\right\| \leq \left\|x^\dagger\right\|.$$

Thus $\left\{x^\delta_{\alpha(\delta)}\right\}_{\delta>0}$ is bounded. By taking a subsequence if necessary, we may assume $x^\delta_{\alpha(\delta)} \rightharpoonup \hat{x}$ as $\delta \to 0$ for some $\hat{x} \in X$. Since A is bounded, it is easy to show that $Ax^\delta_{\alpha(\delta)} \rightharpoonup A\hat{x}$. Using the lower semi-continuity of Hilbert space norms we obtain

$$\|A\hat{x} - y\| \leq \liminf_{\delta\to 0} \left\|Ax^\delta_{\alpha(\delta)} - y^\delta\right\| = 0$$

Thus $A\hat{x} = y$. Recall that $x^\delta_{\alpha(\delta)} \in \mathcal{R}(A^*) \subset \mathcal{N}(A)^\perp$, we have $\hat{x} \in \mathcal{N}(A)^\perp$. Therefore $\hat{x} = x^\dagger$ and hence

$$x^\delta_{\alpha(\delta)} \rightharpoonup x^\dagger \quad \text{and} \quad \left\|x^\dagger\right\| \leq \liminf_{\delta\to 0} \left\|x^\delta_{\alpha(\delta)}\right\|.$$

We therefore obtain

$$x^\delta_{\alpha(\delta)} \rightharpoonup x^\dagger \quad \text{and} \quad \lim_{\delta\to 0} \left\| x^\delta_{\alpha(\delta)} \right\| = \left\| x^\dagger \right\|.$$

Observing that

$$\left\| x^\delta_{\alpha(\delta)} \right\|^2 = \left\| x^\delta_{\alpha(\delta)} - x^\dagger \right\|^2 + \left\| x^\dagger \right\|^2 + 2 \left\langle x^\dagger, x^\delta_{\alpha(\delta)} - x^\dagger \right\rangle.$$

We finally obtain $\lim_{\delta\to 0} \left\| x^\delta_{\alpha(\delta)} - x^\dagger \right\| = 0.$ $\square$

Theorem 3.15

Let $\alpha(\delta)$ be determined by Morozov's discrepancy principle with $\tau > 1$. If $x^\dagger$ satisfies the source condition

$$x^\dagger = (A^*A)^\nu \omega \quad \text{for some } 0 < \nu \le 1/2 \ \text{ and } \ \omega \in \mathcal{N}(A)^\perp,$$

then there is a constant C depending only on τ, q, and ν such that

$$\left\| x^\delta_{\alpha(\delta)} - x^\dagger \right\| \le C \left\| \omega \right\|^{1/(1+2\nu)} \delta^{2\nu/(1+2\nu)}.$$

Proof. We first consider the case that $\left\| y^\delta \right\| \le \tau\delta$. We have

$$\|y\| \le \left\| y^\delta - y \right\| + \left\| y^\delta \right\| \le (\tau + 1)\delta.$$

By using the source condition on $x^\dagger$, and by interpolation inequality ($r = \nu, q = 1/2$), we obtain

$$\left\| x^\dagger \right\|^2 = \left\langle \omega, (A^*A)^\nu x^\dagger \right\rangle \le \|\omega\| \left\| (A^*A)^\nu x^\dagger \right\| \le \|\omega\| \left\| x^\dagger \right\|^{1-2\nu} \left\| Ax^\dagger \right\|^{2\nu}.$$

Therefore

$$\left\|x^\dagger\right\|^{1+2\nu} \leq \|\omega\| \left\|Ax^\dagger\right\|^{2\nu} \leq (1+\tau)^{2\nu} \|\omega\| \, \delta^{2\nu}.$$

Next we consider the case $\left\|y^\delta\right\| > \tau\delta$. By the definition of $x^\delta_{\alpha(\delta)}$ it follows that

$$\begin{aligned}\left\|Ax^\delta_{\alpha(\delta)} - y^\delta\right\|^2 + \alpha(\delta)\left\|x^\delta_{\alpha(\delta)}\right\|^2 &\leq \left\|y - y^\delta\right\|^2 + \alpha(\delta)\left\|x^\dagger\right\|^2 \\ &\leq \delta^2 + \alpha(\delta)\left\|x^\dagger\right\|^2.\end{aligned}$$

Since $\left\|Ax^\delta_{\alpha(\delta)} - y^\delta\right\| = \tau\delta \geq \delta$, we obtain $\left\|x^\delta_{\alpha(\delta)}\right\|^2 \leq \left\|x^\dagger\right\|^2$. Consequently

$$\begin{aligned}\left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^2 &\leq \left\|x^\dagger\right\|^2 - \left\|x^\delta_{\alpha(\delta)}\right\|^2 + \left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^2 \\ &= 2\left\langle x^\dagger, x^\dagger - x^\delta_{\alpha(\delta)}\right\rangle.\end{aligned}$$

By using the source condition on $x^\dagger$ we have

$$\begin{aligned}\left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^2 &\le 2\left\langle \omega, (A^*A)^\nu (x^\delta_{\alpha(\delta)} - x^\dagger)\right\rangle \\ &\le 2\left\|\omega\right\| \left\|(A^*A)^\nu (x^\delta_{\alpha(\delta)} - x^\dagger)\right\|.\end{aligned}$$

Since $0 < \nu \le 1/2$, by using the interpolation inequality (same in previous) we have

$$\left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^2 \le 2\left\|\omega\right\| \left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^{1-2\nu} \left\|(A^*A)^{1/2}(x^\delta_{\alpha(\delta)} - x^\dagger)\right\|^{2\nu}.$$

Observe that $\left\|(A^*A)^{1/2}x\right\|^2 = \left\|Ax\right\|^2$ for all $x \in X$. Therefore

$$\begin{aligned}\left\|x^\delta_{\alpha(\delta)} - x^\dagger\right\|^{1+2\nu} &\le 2\left\|\omega\right\| \left\|(A^*A)^{1/2}(x^\delta_{\alpha(\delta)} - x^\dagger)\right\|^{2\nu} \\ &= 2\left\|\omega\right\| \left\|A(x^\delta_{\alpha(\delta)} - x^\dagger)\right\|^{2\eta} \le 2\left\|\omega\right\|(\tau+1)^{2\nu}\delta^{2\nu}\end{aligned}$$

and we should get the desired result. $\square$

Numerical implementation of Morozov's discrepancy principle:

When $\left\|y^\delta\right\| > \tau\delta$, we need to solve the equation $\varphi(\alpha) = \tau^2\delta^2$, where $\varphi(\alpha) := \left\|Ax_\alpha^\delta - y^\delta\right\|^2$. Recall that $x_\alpha^\delta = (\alpha I + A^*A)^{-1}A^*y^\delta$, we have

$$\varphi(\alpha) = \alpha^2 \left\|(\alpha I + AA^*)^{-1}y^\delta\right\|^2$$

We set $\psi(t) = \varphi(1/t)$. Then

$$\psi(t) = \left\|(I + tAA^*)^{-1}y^\delta\right\|^2 = \left\langle (I + tAA^*)^{-2}y^\delta, y^\delta \right\rangle.$$

Therefore

$$\psi'(t) = -2\left\langle (I + tAA^*)^{-3}AA^*y^\delta, y^\delta \right\rangle < 0,$$
$$\psi''(t) = 6\left\langle (I + tAA^*)^{-4}AA^*y^\delta, AA^*y^\delta \right\rangle > 0$$

Thus φ is a strictly decreasing, convex function on $(0, \infty)$.
Consequently, for any sufficiently small initial guess $t_0 > 0$, the Newton method

$$t_{k+1} = t_k - \frac{\psi(t_k) - \tau^2\delta^2}{\psi'(t_k)}$$

converges monotonically to t_* satisfying $\psi(t_*) = \tau^2\delta^2$ with locally quadratically convergence speed.
Setting $\alpha_k = \frac{1}{t_k}$ and observing $\psi'(t) = -\frac{1}{t^2}\varphi'\left(\frac{1}{t}\right)$. We have

$$\alpha_{k+1} = \frac{\alpha_k^2\varphi'(\alpha_k)}{\alpha_k\varphi'(\alpha_k) + \varphi(\alpha_k) - \tau^2\delta^2}$$

which converges to α_* satisfying $\varphi(\alpha_*) = 0$ monotonically and locally quadratically for any large initial guess α_0.

Pseudo algorithm:

- Input A, y^δ and $\delta > 0$, and pick $\tau > 1$;
- Choose an initial guess $\alpha > 0$, set $\varepsilon := 1$, and $n := 0$;
- While $\varepsilon > 10^{-8}$ do
  1 . $z := (\alpha I + AA^*)^{-1} y^\delta$;
  2 . $x := A^* z$;
  3 . $\varphi := \alpha^2 \|z\|^2$;
  4 . $\varphi' := 2\alpha \|z\|^2 - 2\alpha^2 \langle (\alpha I + AA^*)^{-1} z, z \rangle$;
  5 . $\alpha_1 := \dfrac{\alpha^2 \varphi'}{\alpha\varphi' + \varphi - \tau^2 \delta^2}$;
  6 . $\varepsilon := |\alpha_1 - \alpha|$; $\alpha := \alpha_1$; $n := n + 1$
- end
- Output: n (number of Newton iteration) and x (approximate solution).

Example 10

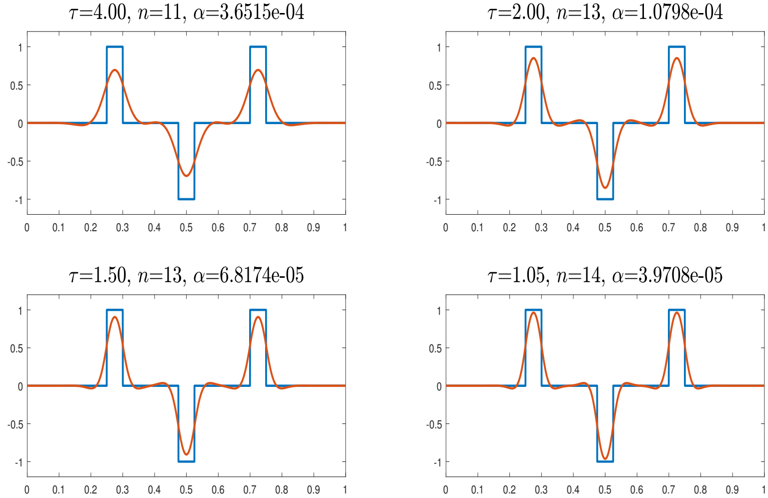


Figure: Redo the second simulation with α determined by the discrepancy principle for different values of τ. We compute α by newton method with initial guess $\alpha = 0.01$, where n denotes the number of Newton steps.

Example 11

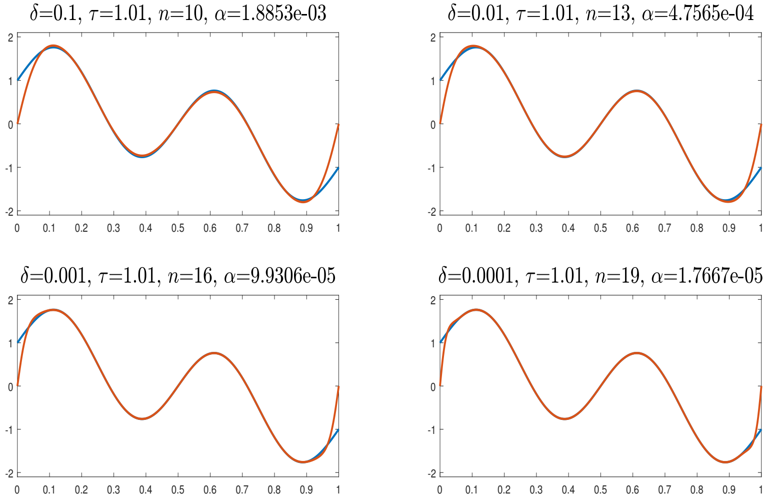


Figure: The solution to be reconstructed is $1 - 2t + \sin(4\pi t)$ under different noise levels δ. The parameter α is determined by the discrepancy principle for $\tau = 1.01$. We compute α by Newton method with initial guess $\alpha = 0.01$, where n denotes the number of Newton steps.

3.2. Landweber iteration

The linear inverse problem $Ax = y$ with noisy data y^δ can be solved formally by finding $x \in X$ to minimize the function

$$x \mapsto \frac{1}{2} \left\| Ax - y^\delta \right\|^2 .$$

Observing the gradient of this function is $A^*(Ax - y^\delta)$. This leads to the gradient method

$$x_{n+1}^\delta = x_n^\delta - \eta A^*(Ax_n - y^\delta), \quad n = 0, 1, \dots,$$

with $x_0^\delta = 0$ and some number $\eta > 0$. This method is called the Landweber iteration.

Example 12

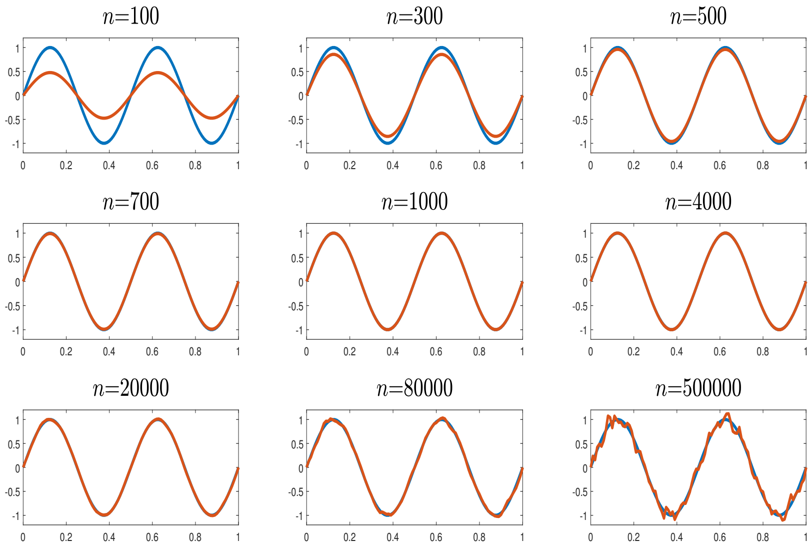


Figure: Numerical reconstruction of the solution $x^\dagger(t) = \sin(4\pi t)$ by Landweber iteration with $\eta = 0.1$, where n denotes the number of iterations.

We first consider the convergence property of the sequence $\{x_n\}$ corresponding to the noise-free case.

Lemma 3.16

If $0 < \eta \leq 2\|A\|^{-2}$, *there holds*

$$\left\|Ax_{n+1}^{\delta} - y\right\| \leq \left\|Ax_n^{\delta} - y\right\|, \quad n = 0, 1, \ldots$$

Proof. By definition of $\{x_n\}$ and $0 < \eta \leq 2\|A\|^{-2}$,

$$\begin{aligned}
&\|Ax_{n+1} - y\|^2 - \|Ax_n - y\|^2 \\
&\qquad = \|Ax_{n+1} - Ax_n\|^2 + 2\langle Ax_{n+1} - Ax_n, Ax_n - y\rangle \\
&\qquad = \eta^2 \|AA^*(Ax_n - y\|^2 - 2\eta\langle AA^*(Ax_n - y), Ax_n - y\rangle \\
&\qquad \leq \eta(\eta\|A\|^2 - 2)\|A^*(Ax_n - y)\|^2 \leq 0.
\end{aligned}$$

This proves the result.

Lemma 3.17

Assume that $0 < \eta \le 2\|A\|^{-2}$ *and let* $c_0 := 2\eta - \eta^2\|A\|^2 > 0$. *Then there hold*

$$\left\|x_{n+1} - x^\dagger\right\| \le \left\|x_n - x^\dagger\right\|, \quad n = 0, 1, \ldots$$

and

$$\sum_{n=0}^{\infty} \|Ax_n - y\|^2 \le \frac{1}{c_0}\left\|x^\dagger\right\|^2.$$

Proof. Observing that

$$\begin{aligned}\|x_{n+1} - x^\dagger\|^2 - \|x_n - x^\dagger\|^2 &= \|x_{n+1} - x_n\|^2 + 2\left\langle x_{n+1} - x_n, x_n - x^\dagger\right\rangle \\ &= \eta^2\|A^*(Ax_n - y)\|^2 - 2\eta\left\langle A^*(Ax_n - y), x_n - x^\dagger\right\rangle \\ &\le (\eta^2\|A\|^2 - 2\eta)\|Ax_n - y\|^2 = -c_0\|Ax_n - y\|^2 \le 0.\end{aligned}$$

This implies the first result. Moreover,

$$c_0 \|Ax_n - y\|^2 \leq \|x_n - x^\dagger\|^2 - \|x_{n+1} - x^\dagger\|^2.$$

This implies for all n that

$$c_0 \sum_{j=0}^{n} \|Ax_j - y\|^2 \leq \|x_0 - x^\dagger\|^2 - \|x_{n+1} - x^\dagger\|^2 \leq \left\|x^\dagger\right\|^2.$$

We therefore obtain the second result. □

Theorem 3.18

If $0 < \eta \leq 2\|A\|^{-2}$, then $\{x_n\}$ converges to $x^\dagger$.

Proof. We first show that $\{x_n\}$ is a Cauchy sequence in X. For any $0 \leq n < m$ we have

$$\|x_n - x_m\|^2 = \left\|x_n - x^\dagger\right\|^2 - \left\|x_m - x^\dagger\right\|^2 - 2\left\langle x_n - x_m, x_m - x^\dagger\right\rangle.$$

From the definition of $\{x_n\}$ it follows that

$$
\begin{aligned}
\left|\left\langle x_n - x_m, x_m - x^\dagger \right\rangle\right| &= \left|\sum_{j=n}^{m-1} \left\langle x_{j+1} - x_j, x_m - x^\dagger \right\rangle\right| \\
&\leq \sum_{j=n}^{m-1} \left|\langle Ax_j - y, Ax_m - y\rangle\right| \\
&\leq \sum_{j=n}^{m-1} \|Ax_j - y\| \, \|Ax_m - y\| \,.
\end{aligned}
$$

Since Lemma 3.16 implies $\|Ax_m - y\| \leq \|Ax_j - y\|$ for $j \leq m$, we obtain

$$
\left|\left\langle x_n - x_m, x_m - x^\dagger \right\rangle\right| \leq \sum_{j=n}^{m-1} \|Ax_j - y\|^2 \,.
$$

Therefore

$$\|x_n - x_m\|^2 \leq \left\|x_n - x^\dagger\right\|^2 - \left\|x_m - x^\dagger\right\|^2 + \sum_{j=n}^{m-1} \|Ax_j - y\|^2$$

Lemma 3.17 implies that $\left\{\left\|x_n - x^\dagger\right\|\right\}$ is monotonically decreasing. It can also be shown that the same sequence is bounded and, hence, convergent. The same lemma also implies that $\sum_{j=n}^{\infty} \|Ax_j - y\|^2 \to 0$ as $n \to \infty$, we therefore obtain

$$\lim_{n,m\to\infty} \|x_n - x_m\| = 0.$$

Thus $\{x_n\}$ is a Cauchy sequence and hence $x_n \to x_*$ as $n \to \infty$. Since Lemma 3.17 implies also $\|Ax_n - y\| \to 0$ as $n \to \infty$, we have $Ax_* = y$. By definition $x_n \in \mathcal{R}(A^*) \subset \mathcal{N}(A)^\perp$. So $x_* \in \mathcal{N}(A)^\perp$ and thus $x_* = x^\dagger$. Therefore $x_n \to x^\dagger$ as $n \to \infty$. $\square$

We next consider the case that the date contains noise. In this situation, Landweber iteration should be terminated in a suitable way. We terminate the iteration by the discrepancy principle

$$\left\|Ax_{n_\delta}^\delta - y^\delta\right\| \le \tau\delta < \left\|Ax_n^\delta - y^\delta\right\|, \quad 0 \le n < n_\delta$$

with $\tau > 1$ and use $x_{n_\delta}^\delta$ to approximate $x^\dagger$.

Lemma 3.19 (Stability estimate)

Assume that $0 < \eta < 2\|A\|^{-2}$ and let $c_1 := \sqrt{2\eta}$. Then for all n there hold

$$\left\|x_n^\delta - x_n\right\| \le c_1 n^{1/2}\delta \quad \textit{and} \quad \left\|A(x_n^\delta - x_n) - y^\delta + y\right\| \le \delta.$$

Proof. Recall that

$$x_n^\delta = x_{n-1}^\delta - \eta A^*(Ax_{n-1}^\delta - y^\delta) \quad \text{and} \quad x_n = x_{n-1} - \eta A^*(Ax_{n-1} - y).$$

Subtraction gives

$$\begin{aligned} x_n^\delta - x_n &= x_{n-1}^\delta - x_{n-1} - \eta A^*(A(x_{n-1}^\delta - x_{n-1}) - y^\delta + y) \\ &= (I - \eta A^*A)(x_{n-1}^\delta - x_{n-1}) + \eta A^*(y^\delta - y). \end{aligned}$$

Recall that $x_0^\delta - x_0 = 0$. Using the above identities inductively gives

$$\begin{aligned} x_n^\delta - x_n &= \eta \sum_{j=0}^{n-1} (I - \eta A^*A)^j A^*(y^\delta - y) \\ &= \eta A^* \sum_{j=0}^{n-1} (I - \eta AA^*)^j (y^\delta - y). \end{aligned}$$

Observing that $\|A^*v\| = \left\|(AA^*)^{1/2}v\right\|$ for all $v \in Y$. We have

$$\left\|x_n^\delta - x_n\right\| \le \eta \sup_{t\in[0,\|A\|^2]} \left|\sqrt{t}\sum_{j=0}^{n-1}(1-\eta t)^j\right| \left\|y^\delta - y\right\|.$$

We set $g_n(t) = \sum_{j=0}^{n-1}(1-\eta t)^j$. Then

$$g_n(t)t = \frac{1}{\eta}\sum_{j=0}^{n-1}(1-\eta t)^j(\eta t - 1 + 1) = \frac{1}{\eta}(1-(1-\eta t)^n).$$

By the condition on η we have $|1-\eta t| \le 1$ for $t \in [0, \|A\|^2]$. Thus

$$0 \le g_n(t) \le n \quad \text{and} \quad 0 \le g_n(t)t \le 2/\eta.$$

Consequently

$$0 \le g_n(t)\sqrt{t} = \sqrt{g_n(t)}\sqrt{g_n(t)t} \le \sqrt{2n/\eta}.$$

Therefore $\left\|x_n^\delta - x_n\right\| \le \sqrt{2\eta n\delta}$ which gives the first estimate.
To derive the second estimate, we use the expression for $x_n^\delta - x_n$ and write $\eta AA^* = I - (I - \eta AA^*)$ to obtain

$$\begin{aligned}
&A(x_n^\delta - x_n) - y^\delta + y \\
&\quad = \sum_{j=0}^{n-1}(I - \eta AA^*)^j(y^\delta - y) - \sum_{j=0}^{n-1}(I - \eta AA^*)^{j+1}(y^\delta - y) - (y^\delta - y) \\
&\quad = -(I - \eta AA^*)^n(y^\delta - y).
\end{aligned}$$

Therefore

$$\left\|A(x_n^\delta - x_n) - y^\delta + y\right\| \le \sup_{t\in[0,\|A\|^2]} |1 - \eta t|^n \left\|y^\delta - y\right\| \le \delta.\square$$

Lemma 3.20

Assume that $0 < \eta < 2\|A\|^{-2}$ *and* $\tau > 1$, *then* n_δ *is finite and*

$$\lim_{\delta \to 0} n_\delta^{1/2}\delta = 0.$$

Proof. From Lemma 3.19 and the definition of n_δ we have for $0 \leq n < n_\delta$ that

$$\|Ax_n - y\| \geq \left\|Ax_n^\delta - y^\delta\right\| - \left\|A(x_n^\delta - x_n) - y^\delta + y\right\| \geq (\tau - 1)\delta.$$

In view of lemma 3.17 we obtain

$$(\tau - 1)^2 n_\delta \delta^2 \leq \sum_{j=0}^{n_\delta - 1} \|Ax_j - y\|^2 \leq \frac{1}{c_0}\left\|x^\dagger\right\|^2$$

This shows that n_δ must be finite.

Next we show $\lim_{\delta\to 0} n_\delta^{1/2}\delta = 0$. We may assume $n_\delta \to \infty$ as $\delta \to 0$. Let $\hat{n}_\delta$ be the smallest integer $\geq n_\delta/2$. Then $n_\delta \to \infty$ and, by Lemma 3.17,

$$(\tau - 1)^2 (n_\delta - \hat{n}_\delta)\, \delta^2 \leq \sum_{j=\hat{n}_\delta}^{n_\delta - 1} \|Ax_j - y\|^2 \to 0$$

as $\delta \to 0$. Since $n_\delta - \hat{n}_\delta \geq n_\delta - (n_\delta/2 + 1) = n_\delta/2 - 1$, we have $n_\delta\delta^2 \to 0$ as $\delta \to 0$. $\square$

Theorem 3.21

Assume that $0 < \eta < 2\|A\|^{-2}$ and let n_δ be the integer determined by the discrepancy principle with $\tau > 1$. Then

$$\lim_{\delta\to 0} \left\| x_{n_\delta}^\delta - x^\dagger \right\| = 0.$$

Proof. Case 1. $n_\delta \to \infty$ as $\delta \to 0$. By using Lemma 3.19 we have

$$\left\|x_{n_\delta}^\delta - x^\dagger\right\| \leq \left\|x_{n_\delta}^\delta - x_{n_\delta}\right\| + \left\|x_{n_\delta} - x^\dagger\right\| \leq c_1 n_\delta^{1/2} n_\delta + \left\|x_{n_\delta} - x^\dagger\right\|$$

In view of Theorem 3.18 and Lemma 3.20 we obtain
$\left\|x_{n_\delta}^\delta - x^\dagger\right\| \to 0$ as $\delta \to 0$.
Case 2. $n_\delta \to n_0$ as $\delta \to 0$ for some finite integer n_0. We may assume $n_\delta = n_0$ for small $\delta > 0$. Since

$$\left\|A x_{n_0}^\delta - y^\delta\right\| \leq \tau\delta \quad \textbf{and} \quad \left\|x_{n_0}^\delta - x_{n_0}\right\| \leq c_1 n_0^{1/2} \delta$$

we obtain $A x_{n_0} = y$. Consequently by the definition of $\{x_n\}$ we have $x_n = x_{n_0}$ for all $n \geq n_0$. Since Theorem 3.18 implies $x_n \to x^\dagger$ as $n \to \infty$, we must have $x_{n_0} = x^\dagger$. Therefore $\left\|x_{n_\delta}^\delta - x^\dagger\right\| \to 0$ as $\delta \to 0$. □

Finally we derive the convergence rates under the source conditions

$$x^\dagger = (A^*A)^\nu \omega \quad \text{with } 0 < \nu < \infty \text{ and } \omega \in \mathcal{N}(A)^\perp.$$

By Lemma 3.19 we have

$$\left\| x_{n_\delta}^\delta - x^\dagger \right\| \leq c_1 n_\delta^{1/2} \delta + \left\| x_{n_\delta} - x^\dagger \right\|.$$

We need to provide an upper bound on n_δ. First we derive an upper bound on $\|Ax_n - y\|$.

Lemma 3.22

Assume that $0 < \eta \leq \|A\|^{-2}$. Under the source condition on $x^\dagger$ there is a constant C_ν depending on η and ν such that

$$\|Ax_n - y\| \leq C_\nu (n+1)^{\nu - 1/2} \|\omega\|, \quad n = 0, 1, \ldots$$

Proof. From the definition of x_n we have

$$Ax_n - y = Ax_{n-1} - y - \eta AA^*(Ax_{n-1} - y) = (I - \eta AA^*)(Ax_{n-1} - y).$$

Since $x_0 = 0$, use an induction hypothesis. This together with $y = Ax^\dagger$ and the source condition on $x^\dagger$ implies

$$\begin{aligned} Ax_n - y &= -(I - \eta AA^*)^n y = -(I - \eta AA^*)^n Ax^\dagger \\ &= -A(I - \eta A^*A)^n (A^*A)^\nu \omega. \end{aligned}$$

Therefore, by using $0 \le \eta \le \|A\|^{-2}$, we obtain

$$\begin{aligned} \|Ax_n - y\| &\le \sup_{t \in [0, \|A\|^2]} |t^{1/2}(1 - \eta t)^n t^\nu| \, \|\omega\| \\ &\le \sup_{s \in [0,1]} |(1 - s)^n s^{\nu + 1/2}| \eta^{-\nu - 1/2} \, \|\omega\| \,. \end{aligned}$$

where we set $s := \eta t$ such that $0 \le s \le 1$.

The function $\varphi(s) = (1-s)^n \, s^{\nu+1/2}$ achieves its maximum on $[0,1]$ at $s_* = \dfrac{\nu+1/2}{n+\nu+1/2}$. Thus

$$\varphi(s) \le \varphi(s_*) \le \left(\frac{\nu+1/2}{n+\nu+1/2}\right)^{\nu+1/2} \le C_\nu (n+1)^{-\nu-1/2}.$$

We therefore obtain the estimate. $\square$

Using Lemma 3.22 we derive that

$$(\tau-1)\delta \le \|Ax_{n_\delta-1} - y\| \le C_\nu n_\delta^{-\nu-1/2} \|\omega\|.$$

This implies that $n_\delta \le C_\nu \left(\|\omega\|/\delta\right)^{2/(1+2\nu)}$ and consequently

$$\left\|x^\delta_{n_\delta} - x^\dagger\right\| \le C_\nu \|\omega\|^{1/(1+2\nu)} \delta^{2\nu/(1+2\nu)} + \left\|x_{n_\delta} - x^\dagger\right\|.$$

Theorem 3.23

Assume that $0 < \eta \leq \|A\|^{-2}$ and let n_δ be the integer determined by the discrepancy principle with $\tau > 1$. If $x^\dagger$ satisfies the source condition

$$x^\dagger = (A^*A)^\nu \omega \quad \textit{with } 0 < \nu < \infty \textit{ and } \omega \in \mathcal{N}(A)^\perp,$$

then there is a constant C_ν depending on ν, τ, and η such that

$$\left\|x_{n_\delta}^\delta - x^\dagger\right\| \leq C_\nu \left\|\omega\right\|^{1/(1+2\nu)} \delta^{2\nu/(1+2\nu)}.$$

Proof. It suffices to estimate $\|x_{n_\delta} - x^\dagger\|$. By the definition of x_n and $y = Ax^\dagger$ we have

$$x_n - x^\dagger = -(1 - \eta A^*A)(x_{n-1} - x^\dagger).$$

By induction and the source condition we have

$$x_n - x^\dagger = -(1 - \eta A^* A)^n x^\dagger = -(A^* A)^\nu (1 - \eta A^* A)^n)\omega.$$

By the interpolation inequality we obtain

$$\begin{aligned}
\left\| x_n - x^\dagger \right\| &\leq \|(1 - \eta A^* A)\omega\|^{\frac{1}{1+2\nu}} \left\| (A^* A)^{\nu+1/2} (1 - \eta A^* A)^n \omega \right\|^{\frac{2\nu}{1+2\nu}} \\
&\leq \|\omega\|^{\frac{1}{1+2\nu}} \left\| (A^* A)^{1/2} (x_n - x^\dagger) \right\|^{\frac{2\nu}{1+2\nu}} \\
&= \|\omega\|^{\frac{1}{1+2\nu}} \|A x_n - y\|^{\frac{2\nu}{1+2\nu}}
\end{aligned}$$

According to Lemma 3.19 and the definition of n_δ we have

$$\|A x_{n_\delta} - y\| \leq \left\| A x_{n_\delta}^\delta - y^\delta \right\| + \left\| A(x_n^\delta - x_{n_\delta}) - y^\delta + y \right\| \leq (\tau + 1)\delta.$$

Therefore $\left\| x_{n_\delta} - x^\dagger \right\| \leq C_\nu \|\omega\|^{\frac{1}{1+2\nu}} \delta^{\frac{2\nu}{1+2\nu}}$. $\square$

Example 13

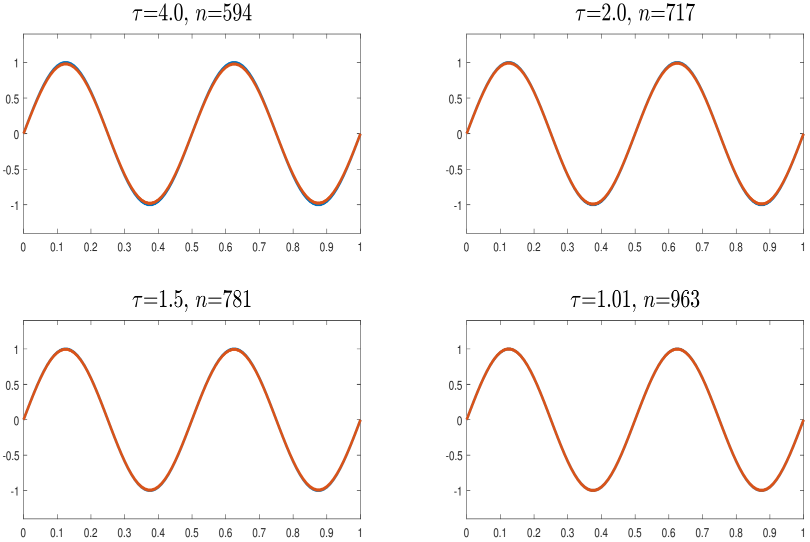


Figure: Redo the second simulation by Landweber iteration by the discrepancy principle with various values of τ. We take $\eta = 0.1$, where n denotes the number of iterations.

Example 14

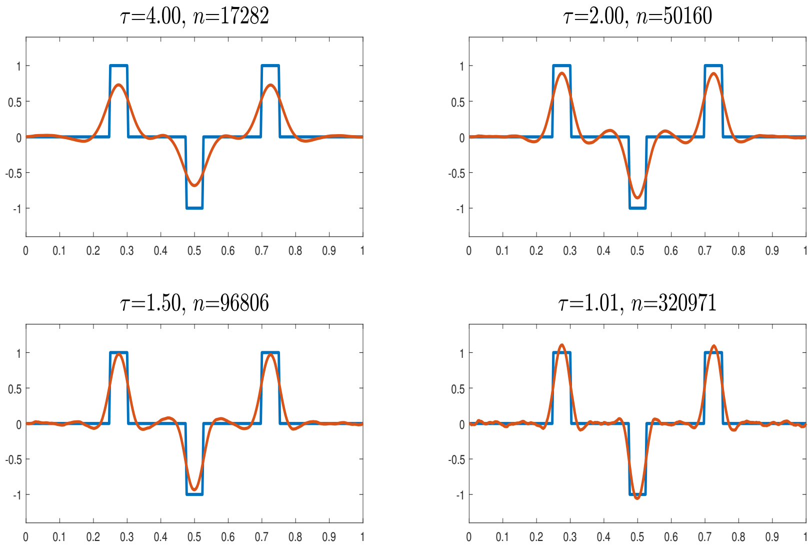


Figure: Redo the second simulation by Landweber iteration by the discrepancy principle with various values of τ. We take $\eta = 0.1$, where n denotes the number of iterations.

Example 15

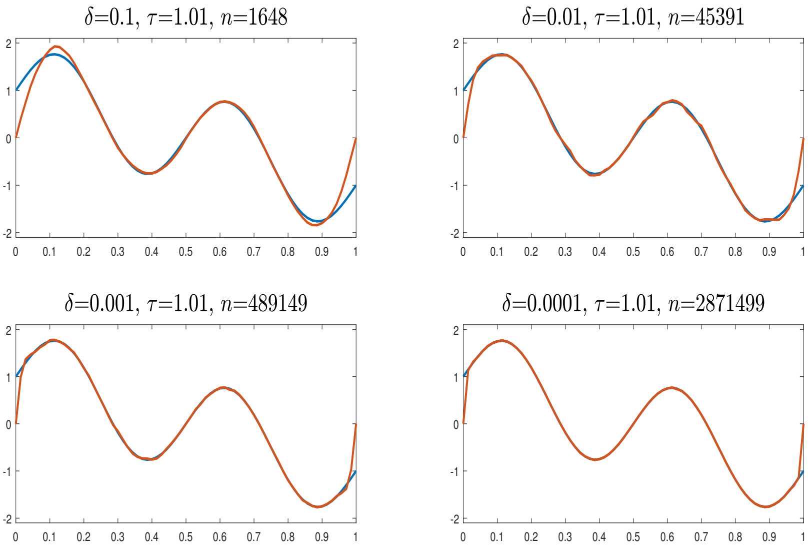


Figure: Reconstruction of the solution $x^\dagger(t) = 1 - 2t + \sin(4\pi t)$ by Landweber iteration with $\eta = 0.1$ terminated by discrepancy principle with $\tau = 1.01$, where n denotes the number of iterations.

Landweber itertion for nonlinear problems

Now we generalize to ill-posed problems of the form

$$F(x) = y$$

where $F : \mathcal{D}(F) \to Y$ with $\mathcal{D}(F) \subset X$. Assuming F is *a nonlinear* operator between Hilbert spaces X and Y, and F has a continuous Fréchet-derivative $F'(x)$, we can define the nonlinear Landweber itertion as

$$x_k^\delta = x_{k-1}^\delta + F'(x_{k-1}^\delta)^*(y^\delta - F(x_{k-1}^\delta)), k \in \mathbb{N}.$$

In the case of noisy data y^δ, we study its convergence via discrepancy principle: the iteration is stopped after $k_* = k_*(\delta, y^\delta)$ steps with

$$\left\| y^\delta - F(x^\delta_{k_*}) \right\| \leq \tau\delta < \left\| y^\delta - F(x^\delta_k) \right\|, \quad 0 \leq k < k_* \tag{4.1}$$

with $t > 0$ appropriately chosen. For nonlinear problems, convergence will *not* be global. In every iteration, $F'(x^\delta_{k-1})$ changes in every iteration, so in general $\{x_k\}$ will not remain within a certain invariant subspace related to a smoothness condition.
Hence, we need to impose some conditions on F.

We assume that our problem is properly scaled:

$$\|F'(x)\| \leq 1, \quad x \in \mathcal{B}_{2\rho}(x_0) \subset \mathcal{D}(F) \tag{4.2}$$

where $\mathcal{B}_{2\rho}(x_0)$ denotes a closed ball of radius 2ρ around x_0. We also need the following local condition (nonlinearity condition):

$$\begin{aligned} &\|F(x) - F(\tilde{x}) - F'(x)(x - \tilde{x})\| \leq \|F(x) - F(\tilde{x})\|, \quad \eta < \frac{1}{2}, \\ &x, \tilde{x} \in \mathcal{B}_{2\rho}(x_0) \in \mathcal{D}(F). \end{aligned} \tag{4.3}$$

From triangle inequality it follows immediately that

$$\frac{1}{1+\eta}\|F'(x)(\tilde{x} - x)\| \leq \|F(\tilde{x}) - F(x)\| \leq \frac{1}{1-\eta}\|F'(x)(\tilde{x} - x)\|.$$

To prove local convergence, we will always assume $F(x) = y$ has a solution in $\mathcal{B}_\rho(x_0)$. The local condition implies the existence of a unique solution of minimal distance to x_0. We call this the x_0-minimum-norm solution, denoted by $x^\dagger$.
For a proof this result see the reference: Kaltenbacher, Barbara / Neubauer, Andreas / Scherzer, Otmar. **Iterative Regularization Methods for Nonlinear Ill-Posed Problems**. 2008

Proposition 4.1

Assume conditions (4.2) and (4.3) hold and that $F(x) = y$ has a solution $x_* \in \mathcal{B}_\rho(x_0)$. If $x_k^\delta \in \mathcal{B}_\rho(x_*)$, a sufficient condition of x_{k+1}^δ to be a better approximation of x_* than x_k^δ is that

$$\left\| y^\delta - F(x_k^\delta) \right\| > 2\frac{1+\eta}{1-2\eta}\delta.$$

Moreover, it then holds that $x_k^\delta, x_{k+1}^\delta \in \mathcal{B}_\rho(x_*) \subset \mathcal{B}_{2\rho}(x_0)$.

Proof. This is Prop. 2.2 in the cited reference. □

This result provides us a suitable choice for τ in (4.1) depending in η:

$$\tau > 2\frac{1+\eta}{1-2\eta} > 2. \tag{4.4}$$

Corollary 4.1

Let the assumptions of Proposition 4.1 hold and let k_* be chosen according to (4.1), (4.4). *Then*

$$k_*(\tau\delta)^2 < \sum_{k=0}^{k_*-1} \left\|y^\delta - F(x_k^\delta)\right\|^2 \leq \frac{\tau}{(1-2\eta)\tau - 2(1+\eta)} \left\|x_0 - x_*\right\|^2 .$$

In particular, if $y^\delta = y$ (i.e. $\delta = 0$, then

$$\sum_{k=0}^{\infty} \left\|y - F(x_k)\right\|^2 < \infty.$$

Proof. This is corollary 2.3 in the cited reference. □

Theorem 4.2

Assume that the conditions (4.2) *and* (4.3) *hold and that $F(x) = y$ is solvable in $\mathcal{B}_\rho(x_0)$. Then the nonlinear Landweber iteration applied to exact data y converges to a solution of $F(x) = y$. If $\mathcal{N}(F'(x^\dagger)) \subset \mathcal{N}(F'(x))$ for all $x \in \mathcal{B}_\rho(x^\dagger)$, then x_k converges to $x^\dagger$ as $k \to \infty$.*

Proof. This is theorem 2.4 in the cited reference. $\square$

Theorem 4.3

Let the assumptions of Theorem 4.2 hold and let $k_* = k_*(\delta, y^\delta)$ *be chosen according to the stopping rule* (4.1), (4.4). *Then the Landweber iterates* $x_{k_*}^\delta$ *converge to a solution of* $F(x) = y$. *If* $\mathcal{N}(F'(x^\dagger)) \subset \mathcal{N}(F'(x))$ *for all* $x \in \mathcal{B}_\rho(x^\dagger)$, *then* $x_{k_*}^\delta$ *converges to* $x^\dagger$ *as* $k \to \infty$.

Proof. This is theorem 2.6 in the cited reference. □

Thank you very much!

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